Initial Labor Market Conditions: Unequal Effects by Parental Income

Diva Astinova
New York University

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This paper examines the unequal impact of initial labor market conditions by parental income and their role in shaping long-term labor market outcomes. Using large-scale Canadian administrative tax data, I analyze the relationship between parental income, aggregate economic conditions, and wage dynamics for new labor market entrants. I find that entering the labor market during an economic downturn leads to significant and persistent earnings losses, with disproportionately larger effects for individuals from low-income families. A 1 p.p. higher initial unemployment rate reduces annual earnings by approximately $2,000 real CAD for all workers over the first decade after graduation, with earnings declining 15 percent more for those from families in the bottom 20% of the income distribution. To explore the underlying mechanisms, I develop a continuous time heterogeneous agent model with financial constraints and a frictional labor market. The model highlights how parental income can act as a safety net: individuals from low-income families are closer to their borrowing constraints and more likely to enter jobs with lower earnings, particularly in the presence of larger search frictions. Recessions amplify this mechanism by reducing job-finding rates in equilibrium, further limiting access to high paying jobs. Counterfactual simulations suggest policies such as targeted grants for graduates entering the labor market could mitigate scarring effects and improve outcomes for disadvantaged workers.

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Author email: diva.astinova[at]nyu.edu

1. Introduction

This paper examines the persistence of intergenerational income inequality, focusing on the interplay between parental income and initial labor market conditions. Parents can provide insurance against idiosyncratic income risks during their children's lifetimes, serving as a financial safety net for their adult children. For financially constrained individuals entering the labor market, this safety net can be particularly crucial to insulate against economic shocks.

At the same time, the conditions under which individuals enter the labor market have long-lasting effects. College students who graduate and enter the labor market in a recession experience average cumulative lifetime earnings losses of 5 percent and decreased wages for up to 10 years (Oreopoulos et al. 2012). Existing research largely treats these initial labor market effects as uniform across individuals, overlooking heterogeneity by socioeconomic status. What is still not well understood is how the effects of initial labor market conditions differ across the parental income distribution. This paper addresses this gap by investigating how initial labor market conditions shape the long-term earnings and employment outcomes across the parental income distribution.

I introduce and establish evidence for a parental insurance mechanism, whereby individuals from lower-income families find themselves closer to the borrowing constraint and more likely to be liquidity constrained compared to their peers with better-off parents. These financial constraints generate different values of unemployment and hence different outside options during the process of labor market search. This heterogeneity initially depends on parental income, and in turn makes individuals with lower-income parents more likely to go into jobs with lower earnings. When initial labor market conditions worsen, such as during recessions, this mechanism is further intensified. Declining job-finding rates and longer unemployment spells further depress outside options, leading to lower earnings in subsequent jobs, especially for individuals from low-income backgrounds.

To analyze this mechanism, this paper uses a rich dataset of Canadian administrative records that links tax files, post-secondary education records and parental income tax data. This dataset allows me to track individuals across their early careers with a high degree of precision and measure how initial labor market conditions affect earnings and employment outcomes across the parental income distribution. I find that initial labor market conditions affect entrants across the parental income distribution, but these effects are unequal. I document that when initial labor market conditions are worse and the initial unemployment rate is rises by 1 percentage point, yearly earnings are on average 10% lower than trend, and these lowered earnings persist for 8 years after graduation.

One of this paper's main contributions is documenting the unequal effects by parental income. I find that labor earnings decline 5 percentage points more for entrants from the poorest 20% of families. These effects are concentrated in the first three years after graduation after which individuals from all family backgrounds suffer similarly lowered earnings for eight years after graduation. The differential effects by parental income are robust to conditioning on a wide range of observables, including gender, degree, major, geography, industry group, institution quality and post-secondary institution fixed effects, with low-income individuals continuing to be especially vulnerable to initial labor market conditions.

Using these empirical findings, I build a heterogeneous-agent model with search frictions, different initial assets and human capital. The core of the model is the interaction between incomplete asset markets and frictional labor search. Individuals are endowed with heterogeneous human capital and initial assets, capturing differences in skill by schooling and different parental safety-nets proportional to parental income. Agents are unable to insure themselves against unemployment shocks - when jobless, they can only dissave to smooth consumption. Unemployed workers frictionally meet vacancies and produce with a technology combining human and physical capital. Upon meeting, the workers and firms bargain over wages, which depend crucially on the worker's outside option in unemployment and therefore current assets. Lower asset holdings, for instance due to a weaker parental safety net, reduce the worker's bargaining position and make the acceptance of lower wages more likely. This mechanism thus maps inequality in the parental safety net into inequalities in labor-market earnings, consistent with the empirical evidence. I use the model to quantify the importance of initial assets and human capital and to assess their roles in amplifying or mitigating the effects of initial labor market conditions.

My primary contribution lies in being the first to document disparities in earnings outcomes for graduates entering the labor market during a recession based on parental income groups. These findings build on an empirical literature studying the effects of graduating in a recession. Additionally, I contribute by developing and solving a quantitative model, offering a theoretical framework to explain and isolate the economic mechanism as well as simulate policies, which complements the predominantly empirical focus of existing research.

Related Literature: The literature finds graduating in a recession has a lasting negative impact on wages. Oreopoulos, von Wachter and Heisz (2012) document that a recession at labor market entry leads to lower labor earnings for up to the next 10 years. The authors examine male college graduates in Canada from 1982 to 1999, and find that graduating in a recession amounts to a decrease in about 8% of cumulative lifetime earnings. Kahn (2010) finds similar persistently lower wages in male college graduates in the US from 1979 to 1989 using the NLSY79. Similar patterns are found by Genda et al. (2010) and Speer (2015) among male high school graduates. The initial negative impact for high school graduates is larger than that for college graduates, and therefore the income losses are more substantial. However, little is known about the differential effects of labor market entry in a recession across the parental income distribution.

To the best of my knowledge, no paper studies the effects of parental income on graduating and entering the labor market in a recession. A growing empirical literature documents the persistence of intergenerational income inequality and the role of parental income in influencing children's lifetime earnings, including Chadwick and Solong (2002) and

Chetty et. al. (2014) in the United States and Corak (2013) in Canada, as well as the role of post-secondary education (see for example Torche, 2011 and Chetty et. al., 2017). The intergenerational relationship of employers between parents and children has recently been documented empirically (Stinson and Wignall, 2014; Kaila et. al., 2024) and is increasing in paternal earnings (Corak and Piraino, 2011), which is expected to widen inequality by family income.

More broadly, family insurance has long been a topic of interest. The literature has demonstrated that parents do insure children by making transfers to less well-off children, as found by Cox (1990), McGarry (2016) and Ameriks et al. (2016). Choi, McGarry and Schoeni (2016) find that extended family income affects individuals' own consumption, and Boar (2021) shows that parents accumulate precautionary savings to insure their children against labor income risk. Moreover, Kaplan (2012) demonstrates moving into the parental home is an important channel to insure against labor market risk, focusing on the effects for male high-school graduates.

The paper is organized as follows. Section 2 describes the data and empirical results, section 3 presents the quantitative model, and section 4 concludes.

2. Empirical Section

We begin by presenting several novel empirical findings. This section first introduces the administrative tax data and defines parental income as used throughout the paper. I then examine the relationship between earnings and parental income, starting with aggregate cohort-level trends, followed by regression analysis on the microdata.

2.1. Data Description

This paper uses Canadian administrative data that combines a panel of individual tax files linked to the tax files of family members, together with a database of post-secondary students in Canada (the T1FF-PSIS). My sample period focuses on students in the years

2006 to 2020. I use the annual unemployment rate at graduation to characterize the initial labor market conditions, following the literature in Oreopoulous et al. (2012). Figure A1 in the appendix shows the variation in the unemployment rate in Canada over time, with recession periods shaded in light gray. The main recession in my sample period is the Great Recession of 2008-09, during which the national unemployment rate for workers aged 15 and up increased by approximately 2 percentage points. In 2015, an oil shock dampened GDP and moderately increased the unemployment rate into 2016, though the economic contraction was not formally classified as a recession. I exclude graduates and new labor market entrants during the Covid recession in 2020 due to the lack of available data in the years following the recession.

2.2. Measure of Parental Income

This paper measures parental income by using the mean parental pre-tax income for each individual observed in the dataset. All income variables are inflation adjusted. This measure of parental income therefore incorporates not only parental labor income, but also income including capital gains, returns on interest and business income, making it a more holistic measure of total parental income and overall wealth. I take the average parental income over at least three years to help ensure we are measuring a good proxy of permanent parental income that is less influenced by temporary income fluctuations. Throughout the paper, I show results by parental income classified into two groups, those in the bottom 20% and top 80% of average real parental income, where the quintile categories are CPI adjusted and defined across all years. My analysis compares individuals who have the same graduation year across groups of parental income. I follow the outcomes of these individuals over years after graduation. The intergenerational income inequality literature commonly analyses parental income in quintiles, with specific interest on individuals from the bottom income quintile. The bottom 20% quintile in my sample is a good representation of individuals from low-income backgrounds and is in line with parental income at or below the Canadian federal poverty threshold. The bottom 20% group also roughly corresponds to poor and liquidity constrained individuals, who

Kaplan and Violante (2014) refer to as the poor hand-to-mouth. In robustness checks, I also run my analysis using parental income groups defined on mean parental income before graduation, five quintile groups as well as using continuous parental income. The results are largely similar.

2.3. Cohort Level Results

I begin by presenting aggregate labor outcomes for graduation cohorts, analyzing the relationship between parental income and the unemployment rate (UR_{c0}) at the time of graduation for graduating cohort c. Figure 1 below shows average labor earnings by group in real Canadian dollars, on the y-axis, plotted against years after graduation on the x-axis. This figure presents raw group means by graduation cohort, parental income group, and years after graduation without conditioning on covariates and includes individuals with zero earnings. The year of graduation is denoted at time zero with a dashed line. Each subplot corresponds to a parental income group. In the figure below, we can see individuals whose parents are in the bottom 20% of parental pre-tax income in the left subplot, and individuals whose parents are in the top 80% of the parental income distribution in the right subplot. Each line denotes a graduating cohort that is colored according to the annual total unemployment rate at graduation, with light blue representing graduating into a good labor market with low unemployment, and orange denoting a more difficult initial labor market with a high unemployment rate.

We observe persistent and long-lasting differences in labor earnings by initial aggregate conditions. For all levels of parental income, Figure 1 shows that individuals who graduate and enter the labor market in better initial conditions, in blue, have higher earnings on average for approximately eight to ten years compared to those who graduate into a worse labor market. This finding is in line with Oreopoulos et. al (2012), who document lasting scarring effects for male college graduates in the 1980s and 1990s. The persistence of initial labor market conditions is surprising, as the effects on mean wages far outlast the duration of the recession, which ended in 2009, and wage effects remain even after

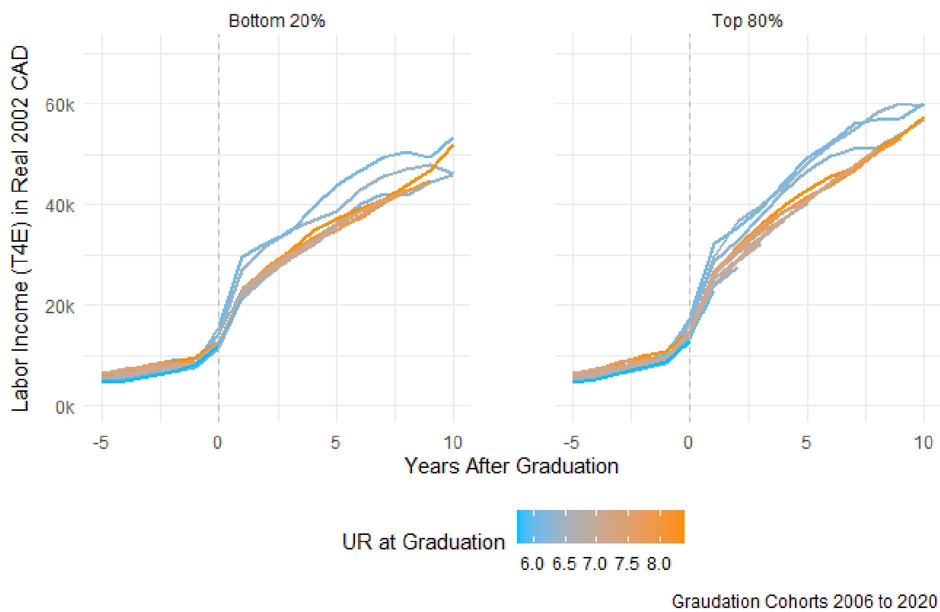


FIGURE 1. Labor Earnings by Parental Income Group and Initial Labor Market Conditions

Note: Each line plots average yearly labor earnings by graduation cohort and parental income group in the years surrounding graduation for 2006 to 2020 post-secondary graduates. Real labor earnings (T4E) in 2002 CAD include zero earnings. The panel on the left shows average earnings for graduates from the bottom 20% of parental income and the panel on the right shows earnings for the top 80% of parental income. See text for detailed description of parental income grouping. Average earnings are colored according to the annual unemployment rate of the graduation year. Source: Canadian T1FF-PSIS microdata.

elevated unemployment rates subside by 2013.

Moreover, we can also see differential effects of earnings across the parental income distribution in any state of the world. In Figure 1, we observe that labor earnings have different profiles by parental income groups irrespective of initial labor market conditions, with individuals from low-income backgrounds having both a lower level and a flatter slope of earnings on average. Because the figure reports labor earnings only (excluding parental transfers), these patterns are not mechanically driven by direct financial support from parents. I account for these average differences across parental-income groups in the micro-level regressions in the following section.

2.4. Regression Specification

To further understand the relationship between initial labor market conditions and parental income, I analyze the mircrodata via regression analysis. I focus on individuals with at least three years of non-missing parental income to proxy for permanent income and, to the best of my ability, and reduce effects of temporary parental income fluctuations. I concentrate on individuals who are aged 26 and younger at the time of their first graduation as they are less likely to have extensive work experience prior to college. My sample contains approximately 1.2 million observations and 200 thousand unique individuals over time who graduated from post-secondary in the sample period. Post-secondary education is defined as education following secondary school, and therefore after high school. My sample therefore includes individuals who complete different degree types, such as one to two year associate degrees and certificates at colleges, as well as those who finish bachelor's degrees and advanced degrees including masters' degrees at universities.

I run the following regression specification dynamically over ten years after graduation, denoted by $j = t - c$, where c is the graduation year and t is the calendar year:

$$
\begin{aligned}
y_{ctpi} = \alpha + & \underbrace{\sum_{j=0}^{10} \beta_{p,j=(t-c)} \text{Parental Inc Group}_p \times \mathbb{1}_{j=t-c}}_{\text{Average effect of parental income group on individual earnings}} \\
& + \underbrace{\sum_{j=0}^{10} \gamma_{0,j=(t-c)} UR_{c0} \times \mathbb{1}_{j=t-c}}_{\text{Average effect of initial UR rate}} + X_i'\delta + \phi_t \\
& + \underbrace{\sum_{j=0}^{10} \gamma_{p,j=(t-c)} UR_{c0} \times \text{Parental Inc Group}_p \times \mathbb{1}_{j=t-c}}_{\text{Additional effect of having parental income group } p} + u_{ctpi}
\end{aligned}
\tag{1}
$$

In my main specification, the dependent variable y_{ctpi} denotes the individual i's real labor earnings in CPI adjusted Canadian dollars, including zeros if the individual is has no labor earnings. I denote the graduation cohort year by c, calendar year t and parental income group p. As described above, the main specification defines parental income group (Parental Inc Group) as a categorical variable for category of parental income p. In some specifications, the dependent variable is restricted to positive labor earnings, and I can therefore look at log earnings. I run further robustness with positive labor earnings above ten thousand 2002 real Canadian dollars, which approximates one year of full-time minimum wage earnings throughout my sample. The effect over time is denoted by $j = (t - c)$, the number of years after graduation.

My main variables of interest are the mean effect of parental income on earnings over time, captured by $\beta_{p,j}$, and the average effect of the initial unemployment rate on earnings in the years following graduation by parental income group p. The average effect of the initial unemployment rate at graduation UR_{c0}, is denoted by $\gamma_{0,j}$ and captures the effect for the reference parental income group, which is the group of parents in the bottom 20% of the parental income distribution. The interaction term $\gamma_{p,j}$ is the additional affect of the initial unemployment rate at graduation for individuals whose parents are in the top 80% of the parental income distribution. This interaction term $\gamma_{p,j}$ is a key parameter of interest; if it is significantly different than zero, then there is evidence of that initial labor market conditions affect entrants differently depending on parental income. With the reference group being those from the lowest socio-economic background, a negative interaction term $\gamma_{p,j} < 0$ signifies that individuals with richer parents have lower earnings when the unemployment rate at graduation UR_{c0} is one percentage point higher. Conversely, if the interaction term is positive, $\gamma_{p,j} > 0$, individuals from lower-income family backgrounds fare worse and have lower earnings than their peers from richer backgrounds when the initial unemployment rate increases.

The regression specification conditions on a number of co-variates. In the baseline specification, I condition on the individual's current age as well as year fixed effects. I progressively condition on additional covariates including gender, currently being enrolled in post-secondary, the degree type, and broad major grouping. I also use information on the individual's current principal industry of employment at the two-digit NAICS level. Degree type is classified by the International Standard Classification of Education (ISCED), which captures differences in post-secondary degrees including between a two-year degree, bachelors' degree or equivalent, masters' degree etc. Major groupings in the PSIS has several available variables categorizing field of study by the Classification of Instructional Programs (CIP) Canada 2016 version, including a primary grouping code (CIPPG), a 2-digit code, and a 4-digit code. I condition on the primary grouping code (CIPPG) in the main results, and have run robustness checks using more detailed 2-digit and 4-digit CIP major codes. In robustness checks, I condition on observables including marital status, geography at the fine census division level, province, and local population size, which is a good measure of the size of the local labor market and labor market opportunities. I also condition on measures of institution quality, namely top Canadian universities, as well as post-secondary institution fixed effects.

2.5. Regression Results

I find that individuals who graduate with a 1 percentage point higher unemployment rate have persistent earnings declines of approximately $1700 real 2002 CAD per year, equivalent to nearly $2700 in 2024 Canadian dollars. This result is robust to numerous control variables including age, gender, degree, type of institution, and major grouping, as shown in Table 1 and Table 2. Moreover, the effect of initial labor market conditions on the path of earnings is highly persistent, with earnings being on average $1700 CAD lower per year for 10 years after graduation, when the unemployment rate at graduation is 1 percentage point higher. That is to say, when comparing two individuals with the same set of observables who are in the same parental income group, those who graduate from post-secondary in a year with a worse labor market and higher unemployment rate earn on average CAD $1700 less labor earnings per year for ten years. These estimates are in line with Oreopolous et. al. (2012), who study the effect of graduating in a recession on earnings for male college graduates in the 1980s and 1990s. In present discounted value, I find that cumulative earning losses amount to over $20,000 real CAD lower earnings in the first ten years after graduation. This effect is therefore economically large.

Figure 2 shows the magnitude of the effect relative to the group average by parental income in the years after graduation. As we can see, individual labor earnings decline by on average 10% relative to the group average over the ten years following graduation. This effect holds for all individuals, and is stronger in the first years after graduation.

Interestingly, the impact of initial conditions on earnings trajectories is heterogeneous by parental income group. This is a novel finding and one of this paper's main contributions. Those in the bottom 20% of the parental income distribution have larger earnings declines compared to when the unemployment rate at graduation is 1 p.p. higher, with their earnings declining by on average $2300 real CAD per year for the first 3 years after graduation. Meanwhile, for individuals with parental income in the top 80% of the parental income distribution, labor earnings decline by on average only $1500 real CAD per year for the first three years after graduation. Thereafter, both groups face similar earnings declines when the initial unemployment rate is lower, regardless of their ranking in the parental income distribution, as shown in Table 1, Table 2 and Figure 3A, as well as in terms of relative earnings in Figure 2. For additional specifications, see Table A1 in the Appendix.

Thus, individuals from the lowest socio-economic class are more vulnerable to initial labor market conditions, namely the unemployment rate at graduation, compared to their peers from middle and upper class backgrounds. This relationship is not purely mechanical, since I am measuring the response to labor earnings that do not include direct transfers from parents to children. My findings hold even when conditioning on observables including gender, degree type, and major grouping. I use additional robustness checks to condition for the top 3 and top 10 institutions, measuring institution quality, as well as observables including geography and living at the parental home, and the results are similar.

TABLE 1. Real Labor Earnings Response to Initial Unemployment Rate UR_{c0} all individuals

	Dependent variable: Real Labor Earnings in 2002 CAD			
	(1)	(2)	(3)	(4)
UR Results *j* Years After Graduation				
Reference Group is Bottom 20% of Parental Income				
UR_{c0} 0 years after graduation	-2,564.20***	-2,533.16***	-2,614.909***	-2,601.44***
	(266.568)	(264.923)	(262.212)	(256.766)
UR_{c0} 1 year after graduation	-2,406.333***	-2,511.994***	-2,660.625***	-2,614.033***
	(269.825)	(268.16)	(265.415)	(259.896)
UR_{c0} 2 years after graduation	-2,155.812***	-2,365.000***	-2,566.850***	-2,407.400***
	(270.301)	(268.636)	(265.886)	(260.355)
UR_{c0} 3 years after graduation	-2,135.527***	-2,128.101***	-2,549.252***	-2,273.418***
	(334.83)	(332.763)	(329.363)	(322.506)
UR_{c0} 5 years after graduation	-2,143.261***	-1,854.507***	-2,522.854***	-2,127.655***
	(394.602)	(392.171)	(388.177)	(380.096)
UR_{c0} 8 years after graduation	-1,570.259***	-1,522.383***	-2,178.521***	-1,929.869***
	(463.406)	(460.545)	(455.843)	(446.349)
UR_{c0} 10 years after graduation	243.156	56.43	-588.499	-361.678
	(479.325)	(476.368)	(471.5)	(461.681)
Age	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y
Sex	Y	Y	Y	Y
In School		Y	Y	Y
Degree Type			Y	Y
College or Uni				Y
Major				Y
R^2	0.167	0.177	0.194	0.227
Adjusted R^2	0.167	0.177	0.194	0.227

Note: * p<0.1; ** p<0.05; *** p<0.01

Note: Average effect of 1 p.p. increase in the initial unemployment rate UR_{c0} in graduation year c on all individual real labor earnings from zero to ten years after graduation. Estimates capture effect of initial UR for all individuals, corresponding to $\gamma_{0,j}$ in regression specification equation 1. Labor earnings are in 2002 CAD and include zero earnings. Individual controls include age, gender, enrolled in post-secondary schooling indicator, post-secondary degree type, two year college or four year university, and major. Standard errors in parentheses. Source: Canadian T1FF-PSIS.

TABLE 2. Real Labor Earnings Response to Initial Unemployment Rate UR_{c0}, differential effect of top 80% compared to bottom 20% of parental income

	Dependent variable: Real Labor Earnings in 2002 CAD			
	(1)	(2)	(3)	(4)
UR x Parental Income Group Results j Years After Graduation				
Interaction of UR_{c0} x Parental Income Group p: Compared to reference group in Bottom 20% of Parental Income				
UR_{c0} × Top 80% Parental Income, 0 years after graduation	483.262*	540.715**	546.344**	440.186*
	(274.617)	(272.922)	(270.128)	(264.503)
UR_{c0} × Top 80% Parental Income, 1 year after graduation	901.228***	695.978**	692.424**	640.292**
	(280.975)	(279.244)	(276.384)	(270.628)
UR_{c0} × Top 80% Parental Income, 2 years after graduation	1,001.086***	828.098***	819.889***	752.918***
	(283.264)	(281.518)	(278.635)	(272.832)
UR_{c0} × Top 80% Parental Income, 3 years after graduation	717.560**	555.255	600.813*	617.670*
	(353.209)	(351.03)	(347.436)	(340.199)
UR_{c0} × Top 80% Parental Income, 5 years after graduation	28.508	-22.738	136.114	216.641
	(418.746)	(416.161)	(411.9)	(403.319)
UR_{c0} × Top 80% Parental Income, 8 years after graduation	-547.585	-567.683	-264.921	-183.363
	(492.351)	(489.311)	(484.3)	(474.21)
Age	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y
Sex	Y	Y	Y	Y
In School		Y	Y	Y
Degree Type			Y	Y
College or Uni				Y
Major				Y
R^2	0.167	0.177	0.194	0.227
Adjusted R^2	0.167	0.177	0.194	0.227

Note: * p<0.1; ** p<0.05; *** p<0.01

Note: Estimates show the differential effect of a 1 p.p. increase in the initial unemployment rate UR_{c0} in graduation year c on real labor earnings by parental income group, for graduates from the top 80% of parental income compared to those from the bottom 20%. See text for detailed description of parental income grouping. Each coefficient corresponds to the differential effect $\gamma_{p,j}$ j years after graduation in regression specification equation 1. Real labor earnings are in 2002 CAD and include zero earnings. Individual controls include age, gender, reference year fixed effects,enrolled in post-secondary schooling indicator, post-secondary degree type, two year college or four year university, and major field of study. Standard errors in parentheses. Source: Canadian TIFF–PSIS tax records.

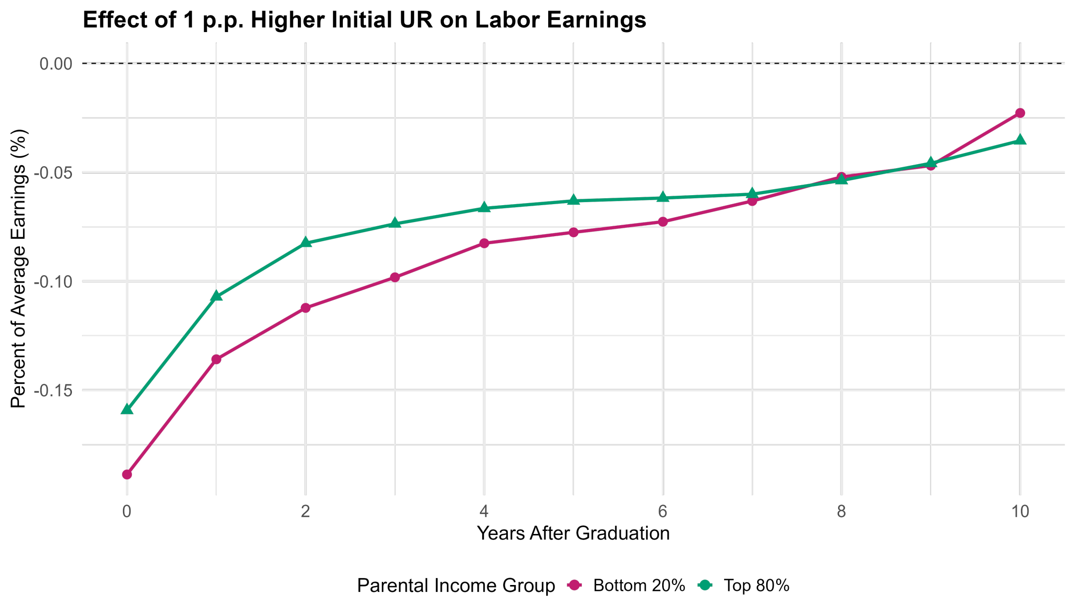
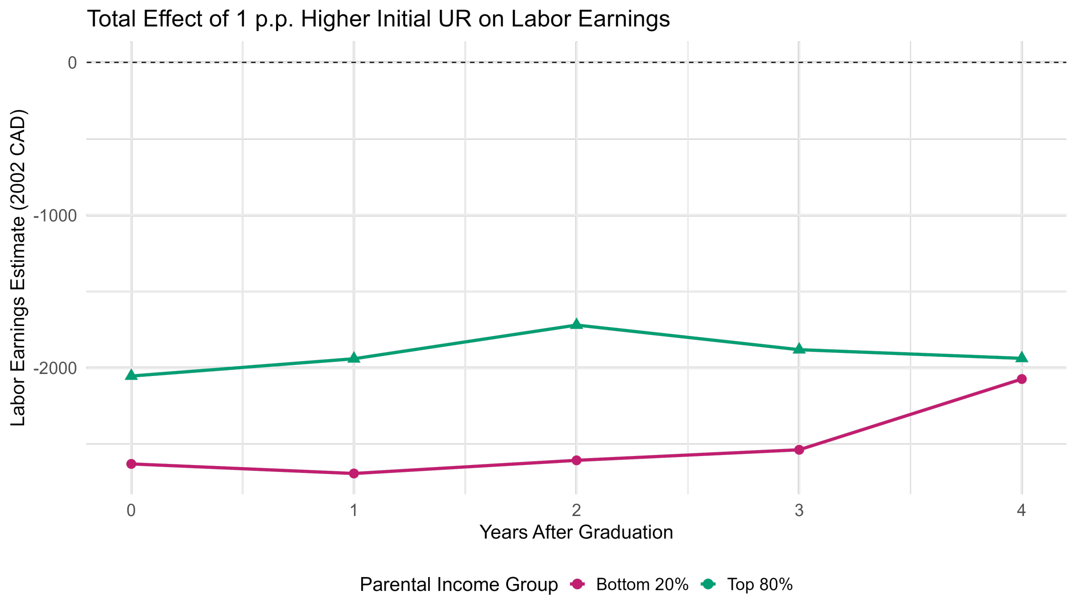


FIGURE 2. Relative effect of Higher Initial Unemployment Rate on Labor Earnings

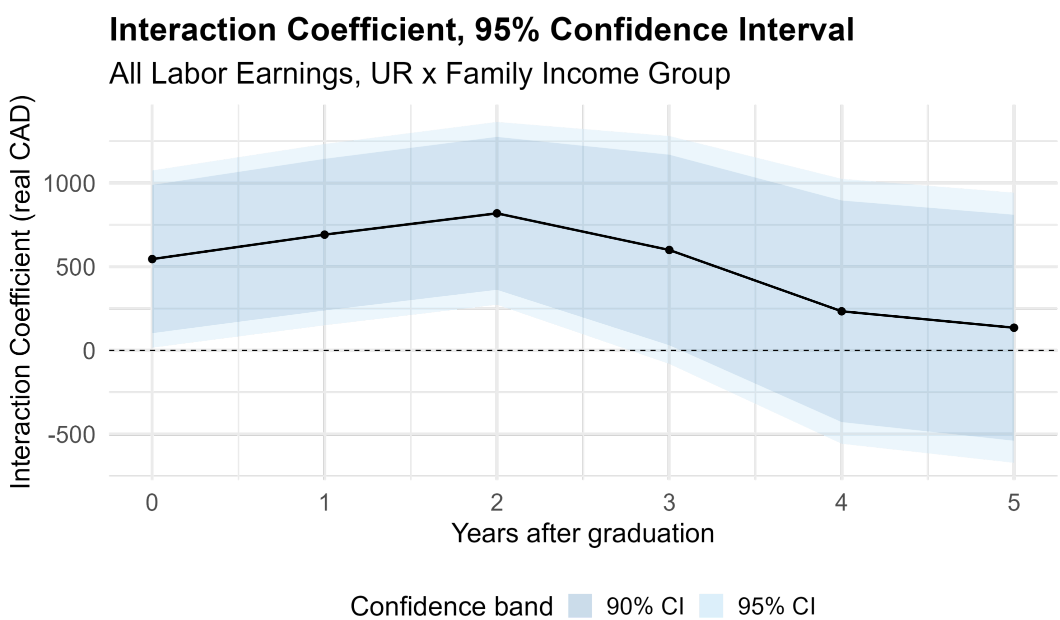
Relative effect on average earnings by parental income group (bottom 20%, top 80%) from a one percentage point increase in the unemployment rate at graduation, relative to own group average earnings, from zero to ten years after graduation. Conditioning on age, gender, year fixed effects, type of post-secondary degree. Levels of earnings including zero earnings. Source: T1FF-PSIS.

These findings are consistent with parents acting as a financial safety net. Individuals from higher and middle income families are further from the borrowing constraint, and are more able to select jobs with higher earnings in the first three years after graduation. In comparison, those individuals from low income families lack the liquidity to do so, and are more pressed to take whichever job is available as they are closer to the borrowing constraint, leading to on average lower earnings. During an economic downturn, particularly the 2008-09 Great Recession, employment becomes harder to find, further amplifying the impact of liquidity constraints and labor market conditions and widening differences by parental income group.

The average earnings of both groups eventually recover to their long run trend. Those from the bottom 20% of the parental income distribution recover slightly faster in approximately nine years after graduation, while those in the top 80% of the parental income distribution take on average ten to eleven years after graduation to return to the long-run trend. This is in line with the observation that those from the bottom 20% of the parental income



A. Level effect on average earnings by parental income group with a one percentage point increase in the unemployment rate at graduation, from zero to four years after graduation. Conditioning on age, gender, year fixed effects, post-secondary degree. All levels of earnings including zero earnings. Source: T1FF-PSIS



B. Confidence intervals of interaction term of parental income group and effect of initial unemployment rate at graduation.

FIGURE 3. Differential effect of initial unemployment rate by parental income group

distribution have a lower average trend of labor earnings compared with individuals from wealthier parents, and therefore their average earnings catch up slightly faster.

Looking at individuals with positive labor earnings only who are in the workforce, we can analyze the effect of initial labor market conditions on log earnings, log y_{ctpi}. Table A2 in the Appendix documents the results. When conditioning on individual positive earnings, individuals with worse initial labor market conditions likewise have a significant and persistent decrease in their earnings. On average, I find that individuals who graduate and enter the labor market with a 1 percentage point higher unemployment rate have 3 percent lower earnings each year for six years following graduation. This persistent result is robust to controls for age, gender, being in school, degree and institution type as well as major and year fixed effects.

Once again, initial labor market conditions have a different impact on earnings by parental income group. Similarly to the results shown in levels, individuals in the bottom 20% of the parental income distribution are more affected by the initial unemployment rate compared to their peers in the top 80%. This heterogeneous effect is concentrated in the first two years after graduation, after which both groups experience similar declines in earnings. Workers from the poorest 20% families have on average approximately 5.5% lower earnings in the year of graduation, and 3.5% lower earnings one year after graduation, all else equal. For workers from the middle and top of the family income distribution, earnings decrease less relative to their trend, with earnings falling on average only 2.5% in the graduation year and 1.5% in the year after graduation when conditioning for the same observables including age, gender, degree type, major group and two digit NAICS industry of employment.

To approximate full-time workers, I focus on individuals whose annual earnings exceed the real full-time minimum wage, which is approximately CAD $10,000 in 2002 constant dollars. Consistent with earlier findings, earnings for these proxied full-time workers are significantly influenced by initial labor market conditions and vary across parental income groups. The results are shown in Table A2 and in relative terms in Figure 4. A

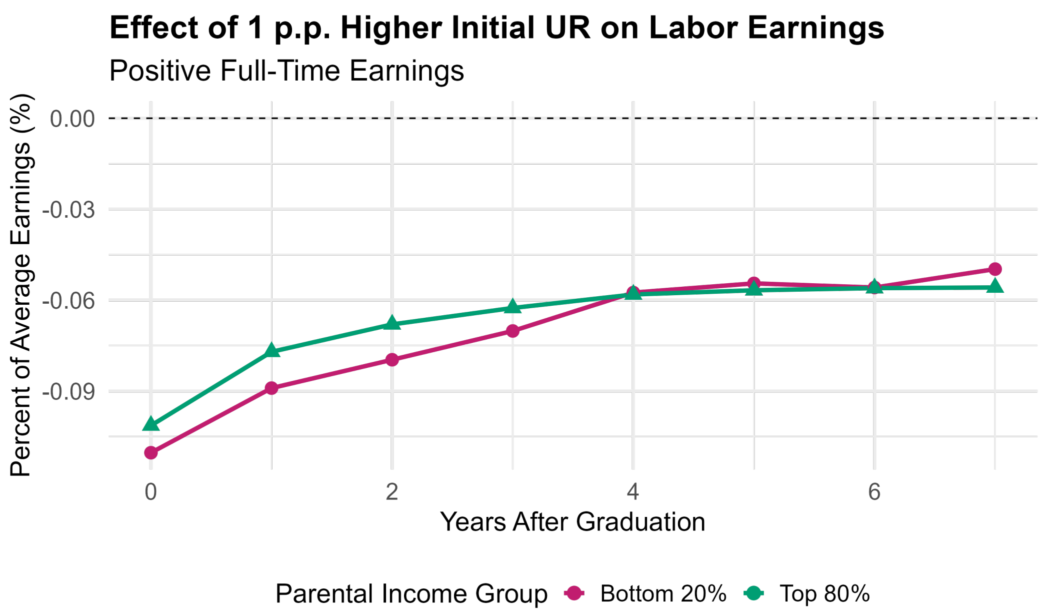


FIGURE 4. Relative Effect of Higher Initial Unemployment Rate on Full-Time Earnings

Relative effect on average full-time earnings by parental income group with a one percentage point increase in the unemployment rate at graduation, relative to own group average earnings, from zero to seven years after graduation. Conditioning on age, gender, year fixed effects, post-secondary degree. Full-time annual earnings levels above full-time minimum wage threshold. Source: T1FF-PSIS

one percentage point increase in the unemployment rate at the time of graduation is associated with an average annual decline in labor earnings of 2% per year over the first seven years post-graduation, holding other factors constant. These effects are both highly persistent and economically significant, underscoring the substantial scarring impact of adverse labor market conditions at graduation on long-term earnings.

The effects on earnings vary significantly by parental income group. Individuals from the bottom quintile of the parental income distribution experience larger earnings declines relative to trend compared to their peers from middle- and upper-income backgrounds. Earnings for those in the bottom 20% of the parental income distribution decline by approximately 3.5%, while the corresponding decline for those in the top 80% is only 1.5%. These differences remain robust when controlling for observable characteristics as shown in Table A3, including gender, degree, major, age, and broad industry grouping at the two-digit NAICS level, ensuring comparability across individuals. Notably, this differential effect persists for up to three years after graduation, after which both groups experience similarly lowered earnings for seven years after graduation when entering a worse initial labor market.

Together these results underscore that individuals from different parental income groups experience different effects of initial labor market conditions. The early post-graduation years, which are critical for career development, reveal significantly different responses to the initial unemployment rate by parental income group, particularly during the first three years following graduation. Workers from low-income families are particularly vulnerable to adverse initial labor market conditions, as they often lack a parental safety net and are more likely to face borrowing constraints. Consequently, they are more inclined to accept available employment out of financial necessity, and are more likely to sort into low earnings jobs, despite having similar observables to their peers from more affluent parents.

Together these empirical results reinforce the importance of parental income in explaining earnings differences across groups. In the following section, I develop a structural model to better understand the relationship between parental income and labor earnings dynamics.

3. Model

Section 3 lays out the structure of the model. The model is a continuous-time heterogeneous agent model with incomplete markets and search frictions. The model combines a Bewley-Huggett-Aiyagari (BHA) framework with Diamond-Mortensen-Pissarides (DMP) frictional labor market, in the spirit of Krusell-Mukoyama-Sahin (2010, 2017) and Achdou et. al. (2022). Individuals face borrowing constraints which increase the reliance on initial parental wealth and lead to different values of unemployment by assets. Individuals are also endowed with human capital level $h \in \mathcal{H}$ that affects their earnings and employment outcomes. I model parental income as a safety-net for new labor market entrants, where initial assets a_0^p are proportional to parental income. Human capital is time invariant in the model, and we can think of human capital as proportional to post-secondary schooling level. At a given instant, each individual is either unemployed (U) or employed (E), with frictional transitions across labor market sectors $s \in \{U, E\}$. The sectors differ in terms of earnings, as described below. In addition, agents choose the optimal levels of consumption c, and savings a'. The state variables are current assets a, human capital h, current labor market sector s and the aggregate state of the continuous-time economy $\mathcal{Z}$.

3.1. Primitives

Consumer Preferences

The economy has a measure one of consumers, who have CRRA utility with relative risk aversion γ

$$U(c) = \frac{1}{1-\gamma} c^{1-\gamma} \qquad \gamma > 1, \gamma \neq 1$$

$$U(c) = \log(c) \qquad \gamma = 1$$

where c denotes the consumption level.

Individual labor income differs by labor market sector s, as well as by human capital level

h and the aggregate state of the economy $\mathcal{Z}$, similarly to Huckfeldt (2022). Labor supply is assumed to be inelastic. Workers have standard time-additive preferences with discount factor ρ and do not value leisure. There are two labor market sectors: unemployment (U) and employment (E). In unemployment, workers receive income from home production, b^U. Employed workers are paid a wage $\omega(a, h, t, \mathcal{Z})$ which depends on their current asset level a, their human capital h, and aggregate economic conditions $\mathcal{Z}$. This wage schedule is continuously negotiated with the firm via Nash bargaining, as will be described in further detail in section 3.6 below.

Production

There are a large measure of potential firms in the economy, one for each job. A firm without a worker can post a single vacancy at cost ξ. An active firm employs one worker with human capital h and rents capital per worker k_t at each instant t.

Firms produce output $y = \mathcal{Z}F(k, h)$ which depends on aggregate economic conditions $\mathcal{Z}$, capital per worker k, and the worker's human capital h. Production F is increasing and concave in human capital, and increasing and strictly concave in physical capital per worker. In the quantification, I assume the functional form $F(k, h) = h + k^\alpha$ for $\alpha \in (0, 1)$. Output is either consumed, invested or used for vacancy creation, as elaborated below.

3.2. Asset Market

There is only one asset in this economy, capital. Firms rent a stock of capital per worker, k_t, continuously at interest rate r_t and capital depreciates at rate δ_k. The gross real interest rate is given by $R_t = 1 + r_t - \delta_k$. Consumers face a borrowing constraint in this asset market, which in the quantification is assumed to be a no-borrowing condition. That is, consumers can only save at the market rate r_t to smooth consumption. In equilibrium, firms demand capital, and consumers supply capital through their savings.

3.3. Labor Market: Search and Matching

Unemployed workers with mass u_t and vacant firms v_t meet randomly at every instant according to the constant returns to scale Diamond-Mortensen-Pissarides aggregate matching function:

$$m(u_t, v_t; Z) = \chi u_t(Z)^{\zeta} v_t(Z)^{(1-\zeta)}$$

where $u_t(Z)$ denotes the total number of unemployed searchers looking for a job and $v_t(Z)$ is the total number of vacancies posted by vacant firms at instant t.

The job-finding rate p_t and vacancy-filling rate q_t are as follows, and vary endogenously with aggregate productivity Z. The job-finding rate depends on the mass of unemployed workers in the economy, u_t, and the vacancy-filling rate depends on the number of vacant job openings v_t.

$$p_t(Z) = \frac{m(u_t, v_t; Z)}{u_t(Z)}, \qquad q_t(Z) = \frac{m(u_t, v_t; Z)}{v_t(Z)} \tag{2}$$

Both the job-finding and vacancy-filling rates can be expressed as a function of the endogenous labor market tightness θ_t, given by $\theta(Z) = \frac{v(Z)}{u(Z)}$.

$$p_t(\theta_t, Z) = m(1, \theta_t; Z), \qquad q_t(\theta_t, Z) = m(\theta_t^{-1}, 1; Z) \tag{3}$$

Job separations, on the other hand, are exogenous, with an employed worker transitioning to unemployment at constant Poisson rate η.

From the above assumptions, unemployment u_t evolves according to the following expression:

$$\dot{u}_t = \eta \times (1 - u_t) - p_t(Z) \times u_t$$

with employed workers who separate from their job flowing into unemployment, and unemployed individuals who match with a firm and become employed flowing out of unemployment.

3.4. Worker's Problem

Workers can either be employed or unemployed. The following HJB equation characterizes the worker's value function in unemployment. At every instant, an unemployed worker chooses an optimal level of consumption taking into account the flow utility and the effect of saving or dissaving, $\dot{a}$, on their value of unemployment. At rate $p_t(\mathcal{Z})$, the unemployed worker meets a vacancy, in which case their value function changes as they transition to employment. The individual takes into account the effect of the sectoral transition, as well as how the value function of unemployment evolves with time captured in the last term below, when optimally choosing their current consumption.

$$\begin{aligned}\rho V^U(a,h;t,\mathcal{Z}) = \max_c \Bigg\{ u(c) &+ \frac{\partial V^U(a,h;t,\mathcal{Z})}{\partial a}\big(\underbrace{(r_t - \delta^k)a + b^U - c}_{\dot{a}}\big) \\ &+ p_t(\mathcal{Z})\left(V^E(a,h;t,\mathcal{Z}) - V^U(a,h;t,\mathcal{Z})\right) \\ &+ \frac{\partial V^U(a,h;t,\mathcal{Z})}{\partial t}\Bigg\}\end{aligned} \tag{4}$$

Consumers face a borrowing constraint which creates a lower bound on their assets, $\underline{a}$. The following state-constraint boundary condition summarizes the borrowing constraint for unemployed workers.

$$\partial_a V^U(\underline{a},h;t,\mathcal{Z}) \geq u'(b^U + (r_t - \delta^k)\underline{a}) \tag{5}$$

Similarly, the below HBJ equation 6 describes the worker's value function when employed. Once again, the worker chooses consumption optimally taking into consideration the flow utility payoff, the effect of saving and accumulating assets on their value function, the evolution of the value function over time, as well as the evolution of the value function with

with exogenous separation at rate η - where the worker transitions from employment to unemployment. A key difference between the value function of employed and unemployed workers are earnings. When employed, a worker receives earnings in the form of the wage $\omega(a, h; t, \mathcal{Z})$, whereas when unemployed, their earnings are their home production/ unemployment benefit b^U. This difference in earnings is highlighted in the savings/ dissavings term $\dot{a}$ below.

$$\rho V^E(a, h; t, \mathcal{Z}) = \max_c \Bigg\{ u(c) + \frac{\partial V^E(a, h; t, \mathcal{Z})}{\partial a} \big(\underbrace{(r_t - \delta^k)a + \omega(a, h; t, \mathcal{Z}) - c}_{\dot{a}} \big) \\ + \eta \left(V^U(a, h; t, \mathcal{Z}) - V^E(a, h; t, \mathcal{Z}) \right) \\ + \frac{\partial V^E(a, h; t, \mathcal{Z})}{\partial t} \Bigg\} \tag{6}$$

At the borrowing constraint, employed workers have assets denoted by $\underline{a}$ and face the following state-constraint boundary condition:

$$\partial_a V^E(\underline{a}, h; t, \mathcal{Z}) \geq u'(\omega(a, h; t, \mathcal{Z}) + (r_t - \delta^k)\underline{a}) \tag{7}$$

Note that the boundary condition is similar for employed and unemployed workers, with the key difference being that earnings for employed workers come from the wage $\omega(a, h; t, \mathcal{Z})$, while unemployed workers have earnings b^U from home production.

Consumption and Savings Policy Functions

Taking the first order conditions of the worker HJB equations yields workers' policy functions for consumption and savings, denoted by $c(a, h, s; t, \mathcal{Z})$ and $\dot{a}(a, h, s; t, \mathcal{Z})$ respectively for consumers in sector $s \in \{U, E\}$.

We can write the consumption policy function generally for both employed and unemployed workers as follows:

$$c(a, h, s; t, \mathcal{Z}) = (u')^{-1}(\partial_a V(a, h, s; t, \mathcal{Z})) \tag{8}$$

The savings policy function is nearly identical for employed and unemployed workers, with slight differences in the source of earnings as we have seen above. Employed workers earn wage $\omega(a, h; t, \mathcal{Z})$, while unemployed workers receive home production b^U. Thus we can express the savings policy funcion as follows:

$$\dot{a}(a, h, s = E; t, \mathcal{Z}) = \omega(a, h; t, \mathcal{Z}) + (r_t - \delta^k)a - c(a, h, s = E; t, \mathcal{Z}) \tag{9}$$

$$\dot{a}(a, h, s = U; t, \mathcal{Z}) = b^U + (r_t - \delta^k)a - c(a, h, s = U; t, \mathcal{Z})$$

Let $g(a, h, s; t, \mathcal{Z})$ be the distribution of consumers across individual states a, h and sector $s \in \{U, E\}$ conditional on aggregate state $\mathcal{Z}$. The savings policy can then be used in the Kolmogorov forward equation that will characterize the dynamics of these distributions across individual states. I drop the aggregate state $\mathcal{Z}$ from the equations below for ease of notation.

$$\partial_t g(a, h, E, t) = -\partial_a[\dot{a}(a, h, E; t,)g(a, h, E, t)] + p_t g(a, h, U, t) - \eta g(a, h, E, t) \tag{10}$$

$$\partial_t g(a, h, U, t) = -\partial_a[\dot{a}(a, h, U; t,)g(a, h, U, t)] + \eta g(a, h, E, t) - p_t g(a, h, U, t) \tag{11}$$

3.5. Firm's Problem

Firms create jobs, pay wages to employed workers, rent capital from consumers, and produce. As discussed above, output depends on aggregate conditions $\mathcal{Z}$ and is increasing human and physical capital in the firm's production function $F(h, k)$. Wages are negotiated with workers at every instant via Nash bargaining, and the firm optimally chooses the amount of physical capital per worker, k, to rent at every instant according to the real interest rate r_t. Firms discount future profits at the market rate $r_t - \delta^k$. As in the standard

Diamond-Mortensen-Pissarides framework, each firm is a single job that can employ at most one worker, and vacant firms can post a single vacancy. Here J denotes the value of a firm employing one worker, and the value of a vacant firm with no workers is denoted by J_0.

The firm's HJB value function $J(a, h; t, \mathcal{Z})$ when employing a worker with assets a and human capital h is as follows.

$$
\begin{aligned}
(r_t - \delta^k) J(a, h; t, \mathcal{Z}) = \max_k \Big\{ & \mathcal{Z} F(h, k) - \omega(a, h, t, \mathcal{Z}) - r_t \times k \\
& + \frac{\partial J(a, h; t, \mathcal{Z})}{\partial a} \dot{a}(a, h, s = E; t, \mathcal{Z}) \\
& + \eta \big(J_0 - J(a, h; t, \mathcal{Z}) \big) \\
& + \frac{\partial J(a, h; t, \mathcal{Z})}{\partial t} \Big\}
\end{aligned}
\tag{12}
$$

The firm optimally chooses the amount of capital per worker k to rent at every instant, taking into account the effect of k on the firm's profits as well as its continuation value. The value of the firm also depends on the worker's saving choice $\dot{a}$, which the firm takes as given, since it changes the worker's assets and therefore affects future wage bargaining and subsequent wages. The firm also takes into account the possibility of becoming exogenously separated from its worker at Poisson rate η and becoming a vacant firm, as well as how the value of the firm evolves over time.

A potential vacant firm takes into account the distribution of agents across states of the world, given by $g(a, h, s, t, \mathcal{Z})$ with sector $s \in \{U, E\}$, and decides whether or not to enter. Upon entering, a vacant firm J_0 pays a the flow cost ξ, every instant t to post a vacancy and create a job.

The value of the vacant firm J_0 is as follows:

$$(r_t - \delta^k)J_0(t, \mathcal{Z}) = -\xi + q_t(\mathcal{Z}) \int_a \int_h J(a, h; t, \mathcal{Z}) \frac{g(a, h, u; t, \mathcal{Z})}{u_t} dhda + \partial_t J_0(t, \mathcal{Z}) \tag{13}$$

The first term captures the payment of the flow vacancy posting cost ξ, and the second term denotes the expected value of matching and producing with an unemployed worker of assets a and human capital h. At rate $q_t(\mathcal{Z})$, which varies endogenously with the labor market tightness and aggregate economic conditions, posted vacancies are filled and the firm produces following $J(a, h; t, \mathcal{Z})$. When deciding to enter, the potential firm considers the pool of unemployed workers with which it could randomly match, since the worker's human capital h and assets a affect the firm's output and costs (via wages). The expectation of worker type across the unemployment pool is summarized in the marginal distribution $\frac{g(a,h,u;t,\mathcal{Z})}{u_t}$ of unemployed workers over assets and human capital. The value of the vacant firm also evolves over time t, as shown in the last term.

With free entry of firms, firms will post vacancies until the value of any potential entrant equals to zero and we have the following equilibrium condition:

$$\xi = q_t(\mathcal{Z}) \int_a \int_h J(a, h; t, \mathcal{Z}) \frac{g(a, h, u; t, \mathcal{Z})}{u_t} dhda \tag{14}$$

Namely, in equilibrium with free entry, the flow vacancy posting cost ξ is equal to the expected value of meeting an unemployed worker with assets a and human capital h and filling the vacancy.

The matched firm's first-order condition with respect to k implies

$$r_t = \mathcal{Z} F_k(k, h)$$

as in the usual relationship equating the marginal cost of capital to the marginal factor demand for capital per worker.

3.6. Wage Setting

A newly formed job-worker match generates a surplus that must be split between the worker and the firm. As is common in the DMP literature, wages are determined by Nash bargaining between a matched worker and firm. I assume that the firm cannot commit to future wages, and therefore flow wages are bargained at every instant, similarly to Krusell et. al. (2010). Current assets are taken into account in the bargaining because they affect worker and thereby firm's continuation values. The Nash bargaining solves the following problem with maximizer $w = \omega(a, h, \mathcal{Z})$. Here $\nu \in (0, 1)$ denotes the worker's bargaining power. For notational simplicity, I omit the time subscripts.

$$\max_{w}\Big(V^E(w, a, h, \mathcal{Z}) - V^U(a, h, \mathcal{Z})\Big)^{\nu}\Big(J(w, a, h, \mathcal{Z}) - J_0(a, h, \mathcal{Z})\Big)^{1-\nu} \tag{15}$$

The wage depends on worker's asset and human capital level, as well as aggregate economic conditions. Current assets are taken into account in the bargaining because they affect worker and thereby firm's continuation values. This dependence is highlighted by the dependency of V^E and J on w in the above equation. In continuous time, V^E and J are both linear in $\omega(a, h, \mathcal{Z})$, which makes solving the problem computationally more tractable.

3.7. Stationary Equilibrium

3.7.1. Definition of Stationary Equilibrium

I now define a stationary equilibrium in this setting. To simply notation, I will omit the time subscripts, the aggregate economic state $\mathcal{Z}$ and partial derivatives. Recall that the individual states are assets a, human capital h, and sector $s \in \{U, E\}$.

The stationary recursive equilibrium consists of

1. a set of value functions for consumers and firms $\{V^U(a, h), V^E(a, h), J(a, h), J_0\}$
2. a set of policy functions $\{c(a, h, s), \dot{a}(a, h, s)\}$
3. a distribution over assets, human capital, and sectors $g(a, h, s)$;

4. a set of prices $\{r, \omega(a,h)\}$;
5. a capital stock per worker k; and
6. labor market tightness θ

such that:

(i) Consumers are optimizing. Given prices and the labor market tightness, the policy functions $\{c(a,h,s), \dot{a}(a,h,s)\}$ solve the consumer optimization problem given by equations (4), (5) and (6), (7), with the value functions $V^U(a,h)$ and $V^E(a,h)$ for unemployed and employed workers respectively.

(ii) Given the consumers decisions rules and tightness, the distribution $g(a,h,s)$ satisfies the Kolmogorov forward equation (10) and (11).

(iii) Firms are optimizing. Given prices, labor market tightness, and the stationary distribution, k optimally solves the firm's problem (12) with corresponding value functions $J(a,h)$ and J_0.

(iv) There is free entry of firms. Given prices, labor market tightness, and the stationary distribution, J_0 satisfies equation (13) with the measure of vacancies adjusting to meet the condition.

(v) In the labor market, the job-finding and vacancy-filling rates are given by the labor market tightness θ according to (3). The measure of unemployed workers u is given by

$$u = \frac{\eta}{\eta + p(\theta)} = \int_a \int_h g(a,h,u)dhda$$

The measure of vacancies is then given by $v = u\theta$.

(vi) Wages are negotiated at every instant via generalized Nash bargaining between a matched worker and firm following equation (15).

(vii) The asset market clears.

$$\int_a \int_h a[g(a,h,E) + g(a,h,U)]dadh = K$$

The aggregate supply of capital is determined from consumers' savings, while the aggregate demand of capital by firms is K. For symmetric firms, aggregate capital can be expressed in relation to all employed workers' capital per worker $K = (1-u)k$.

(viii) The goods market clears.

$$\int_a \int_h \sum_{s \in \{E,U\}} c(a,h,s)g(a,h,s)dadh + \delta_k K + \xi v = \int_a \int_h \mathcal{Z} F(h,k)g(a,h,E)dadh$$

Firms and home production produce the final good, and output is used for consumption, investment in physical capital, and vacancy creation.

3.8. Estimation

To quantitatively solve the model, I assign standard values for the assigned parameters in Table 3. The risk aversion parameter in the CRRA utility function is set with $\gamma = 1$. Aggregate economic conditions $\mathcal{Z}$ are normalized to one in the stationary baseline. In the baseline, the borrowing limit is set to zero to impose a no-borrowing constraint. The borrowing constraint can be relaxed to allow a fixed level of borrowing $\underline{a}$ for all agents.

TABLE 3. Assigned Parameters

Parameter	Description	Value
γ	Risk aversion	1
$\underline{a}$	Borrowing limit	0
$\mathcal{Z}$	Aggregate economic conditions, baseline	1
ζ	Elasticity of matching wrt unemployment	0.72
ν	Worker's bargaining power	0.72
α	Capital share in production	0.3

TABLE 4. Estimated Parameters

Parameter	Description	Value
δ_k	Depreciation rate of capital	0.022
ρ	Discount factor	0.01
b^u	Home production	0.75
χ	Matching efficiency	1.793
η	Exogenous separation rate	0.1038
h_2/h_1	Human capital productivity ratio	4.00
ξ	Vacancy posting cost	0.464

TABLE 5. Model vs Data Moments

Moment Name	Model	Data	Source
Investment-output ratio	0.188	0.200	Krusell et. al. (2010)
Ann. real rate of return on capital	0.039	0.040	Krusell et. al. (2010)
Home production	$0.366\bar{w}$	$0.4\bar{w}$	Shimer (2005)
Monthly job finding rate	0.453	0.45	Shimer (2005)
Monthly separation rate	0.034	0.034	Shimer (2005)
Skill wage ratio	1.155	1.20	Data
Labor market tightness	1.032	1	Shimer (2005)

The elasticity of the matching function with respect to unemployment is set to $\zeta = 0.72$, and the bargaining parameter ν which captures the worker's bargaining power is set according to the Hosios condition $\nu = \zeta = 0.72$, following Shimer (2005). In the standard DMP framework, the Hosios condition would ensure efficiency. However, with workers' precautionary savings, efficiency is not guaranteed in this setting.

For the remaining parameters shown in Table 4, I jointly estimate the parameter set ψ_p by matching moments. The estimated values minimize the following objective function following Freund (2024):

$$\mathcal{G}(\psi_p) = \sum_j \left(\frac{\hat{m}_j - m_j(\psi_p)}{\frac{1}{2}\|\hat{m}_j\| + \frac{1}{2}\|m_j(\psi_p)\|} \right)^2$$

where $\hat{m}_j$ refers to the empirical moment j, and $m_j(\psi_p)$ denotes its model counterpart.

These moments are jointly estimated, there is a clear mapping between each parameter in Table 4 and its targeted moment in Table 5. I estimate the discount factor ρ, targeting the annual real rate of return of capital as in Krusell et. al. (2010). The depreciation rate of capital is calibrated according to the investment-output ratio.

The value of home production is estimated to match 40% of the average wage when employed, following the literature from Shimer (2005). The matching efficiency χ governs the job-finding rate, as increasing the efficiency in the matching function increases the rate of matches between unemployed workers and vacancies. The exogenous separation rate, on the other hand, is directly disciplined by the monthly separation rate.

The observed skill wage premium in the microdata disciplines the relative efficiency of human capital in production. In the quantification, human capital takes two levels. The lower level of human capital, h_1, corresponds to individuals with less than a bachelor's degree, while the higher level of human capital, h_2, parallels individuals with a bachelor's degree or higher. This two-tier classification follows standard practice in the labor literature, reflecting the sizable earnings gap between bachelor's and above graduates compared to those with some college, such as associate degrees or certificates. Bringing the model to the data, I target the approximately 20% college wage premium I calculate in my sample. This data moment is in line with aggregate Canadian data on the relative earnings premium of bachelor and higher graduates to post-secondary graduates who have attained less than a bachelor's degree.

TABLE 6. Untargeted Model vs Data Moments

Moment Name	Model	Data	Source
Log ratio top/bottom 20% assets	1.678	1.674	Data
Ratio top/bottom 20% assets	5.355	5.333	Data

Finally, as in Shimer (2005), I target a the labor market tightness of one in the data, and discipline the model labor market tightness by adjusting the flow vacancy-posting cost ξ. Intuitively, the higher the cost of posting a vacancy, the fewer firms enter, decreasing the available vacancies v_t and reducing the labor market tightness θ_t.

The model also performs well on non-targeted moments at the heart of the paper's analysis regarding the distribution of parental income, as seen in Table 6. In the data, we compute two measures of dispersion in the asset distribution. The first is the ratio between average asset holding of the top 20% of the equilibrium distribution against the average asset holding of the bottom 20%. The empirical target is the analogous ratio of my measure of parental income between the parental income of the top and bottom 20%. The second measure of asset dispersion is similar as the log ratio between average asset holding of top 20% and bottom 20 %, with the corresponding data moment taken from the log ratio of the parental asset distribution. I compute the model analogues of these objects in terms of the stationary distribution of assets, capturing the intuition that stationary assets correspond proportionally to the parental-income distribution safety net for individuals observed in the empirical data.

3.9. Simulations

Using the calibrated model, I conduct several quantitative exercises. In the first exercise, I simulate a contraction in aggregate conditions $\mathcal{Z}$ to simulate the Great Recession's impact on initial labor market conditions. The model allows me to then decompose these effects by initial human capital and by initial asset levels.

As in the data, I study the effect of initial labor market conditions on the earnings trajecto-

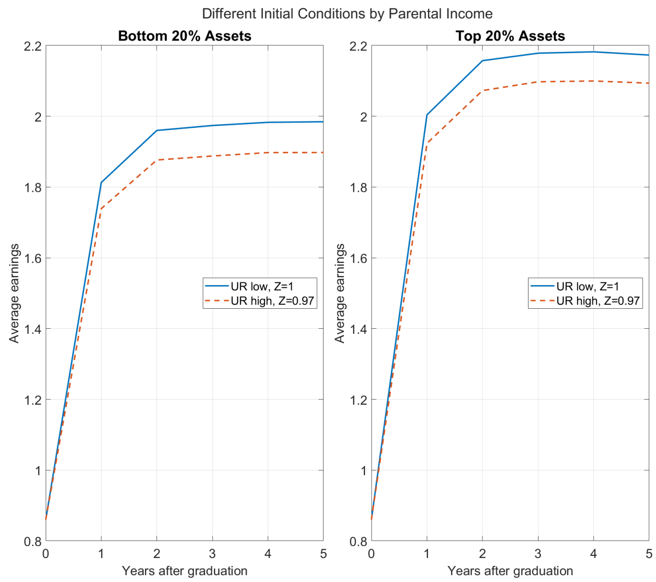


FIGURE 5. Model Moments, Average Earnings by Initial Labor Market Conditions and Parental Income

ries of post-secondary graduates. With the calibrated model, I simulate two steady-state economies. The first is a baseline economy with favorable initial labor market conditions, corresponding to a stationary environment where Z is normalized to one. As in the empirical section, this baseline economy will be denoted in the graph by a solid blue line. The second economy represents a stationary environment with adverse initial labor market conditions, where aggregate productivity Z declines by 3% relative to the baseline. This second counterfactual economy emulates graduating into a labor market with a high unemployment rate, such as during the Great Recession in Canada, and is plotted as an orange dashed line. I simulate a large mass of agents in each economy by parental income group, comparing individuals from low-income families (bottom 20% of the parental income distribution) to those from high-income families in the top 20% of parental income. Average earnings after graduation are plotted in Figure 5 below. Consistent with the data, few graduates are employed full-time at graduation, so I initialize all individuals as unemployed. Over time, unemployed graduates meet vacancies and transition to employment, which raises average earnings quickly in early years. This mirrors the pattern of rising average earnings after graduation seen in the cohort-level data in Figure 1.

When the initial labor market conditions are worse (shown in the orange dashed line) the job-finding rate declines and individuals from all parental backgrounds take longer to find employment. Moreover, recessions lower match surplus, so equilibrium wages fall relative to the baseline.

Second, wages are on average lower for individuals from low-income backgrounds, and with a smaller parental safety net, in any state of the world. Since individuals from low-income parents have very little safety net, they find themselves starting closer to the borrowing constraint. The nonlinearities near the borrowing limit cause graduates from low-income backgrounds to have much lower values of unemployment, which substantially reduces their outside option in the wage bargaining when they do match with a firm. Consequently, their initial earnings are lower than those of peers from more affluent families.

Both these patterns reflect what we see in the data. Figure 6 shows the empirical analogue to the simulations, plotting average earnings by parental income group under baseline and adverse labor market conditions. In the figure, low initial UR corresponds to graduation cohorts from 2006 to 2008, high initial UR corresponds to graduation cohorts from 2009 to 2011. The model does a good job of qualitatively matching the initial wage patterns by parental income and initial labor market conditions, and thus provides qualitative support for the parental-safety net mechanism. The present exercise intentionally relies on steady states to isolate the mechanism; transitional dynamics are quantified by introducing an unexpected MIT shock in section 3.10, which delivers the wage mean reversion observed after recessions.

Moreover, in the model human capital is time-invariant, which abstracts from on-the-job learning that can further increase workers' productivity and earnings. This is a deliberate simplification to sharpen the role of borrowing constraints and search. As a result, the model does not generate life-cycle growth in earnings. Allowing for human capital to accumulate through on the job earnings is a natural extension. Since the main focus of the paper is on the first years following graduation, where time variation in human

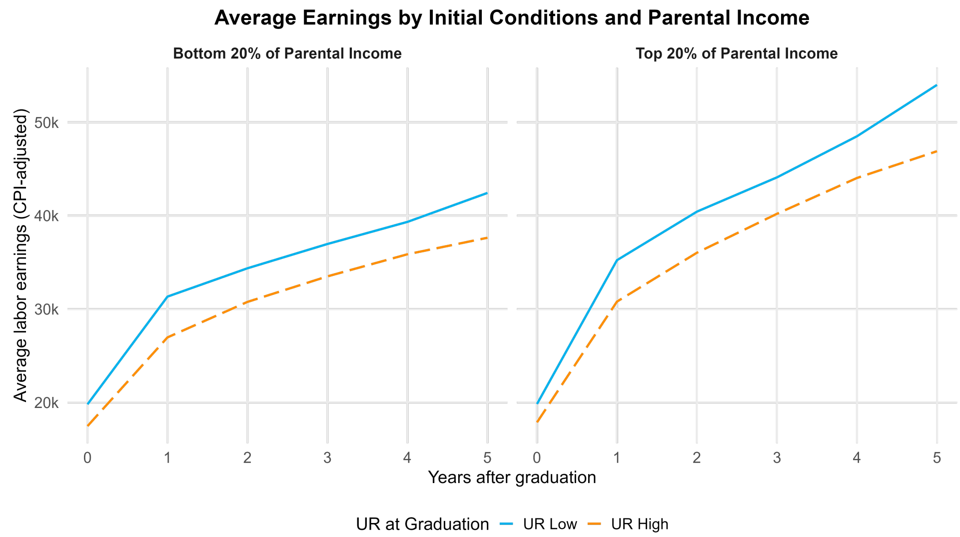


FIGURE 6. Aggregate Data Moments, Average Earnings by Initial Conditions & Parental Income

Note: Average yearly labor earnings by graduation cohort, initial labor market conditions, and parental income group in the years after graduation for 2006 to 2011 post-secondary graduates. Low unemployment rate averages graduating cohorts 2006-2008, while high unemployment rate averages graduating cohorts 2009-2011, corresponding to the Great Recession in Canada. Real labor earnings (T4E) in 2002 CAD include zero earnings. The panel on the left shows average earnings for graduates from the bottom 20% of parental income and the panel on the right shows earnings for the top 80% of parental income. See text for detailed description of parental income grouping. Average earnings are colored according to the annual unemployment rate of the graduation year. Source: Canadian T1FF-PSIS microdata.

capital is limited, this stationary exercise serves as a clean and informative benchmark for evaluating the model mechanism against the data.

3.10. Transition Dynamics

In this section, I study the transition dynamics after an unexpected, transitory (MIT) shock to aggregate productivity $\mathcal{Z}$. This exercise can speak to the recovery across the asset distribution following a shock in aggregate economic conditions in Canada during the Great Recession. The economy begins in its stationary equilibrium with $\mathcal{Z} = 1$. At time zero, there is a one-time and unanticipated fall to aggregate productivity by 3% to $\mathcal{Z}_0 = 0.97$, after which the economy gradually reverts back to the initial steady state along a deterministic path. Agents anticipate that $\mathcal{Z}$ returns to its stationary equilibrium value at exponential rate ζ_z, where $\dot{\mathcal{Z}} = \zeta_z(\mathcal{Z}_t - 1)$.

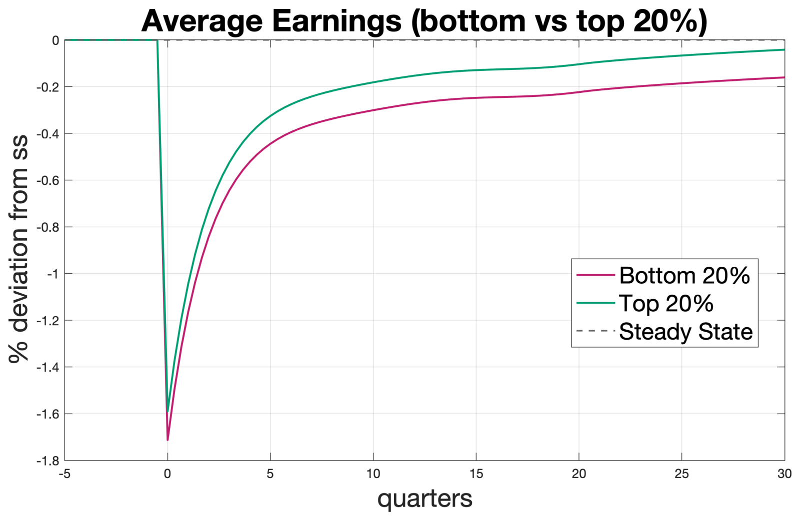


FIGURE 7. Transition dynamics of average earnings after a one-time 3% decrease in aggregate productivity $\mathcal{Z}$ (MIT shock), in percent deviation from steady state.

To assess distributional effects, I track average earnings for individuals in the bottom 20% and the top 20% of the cross-sectional asset distribution. I zoom in on average earnings in the first 30 quarters after the shock to show an effect with a similar scale to the empirical section. In the figure, the bottom 20% of the cross-sectional asset distribution is denoted in magenta, while the top 20% of assets is denoted in green.

On impact, average earnings fall below their steady-state levels for both groups, with a larger decline for low-asset individuals, who are pushed closer to the borrowing constraint. Over time, average earnings return to their stationary levels, but remain depressed for many periods. The recovery is slower for low-asset households, widening the gap with high-asset households before it gradually closes. These dynamics arise because the negative productivity shock strengthens precautionary saving, pushes low-asset agents nearer to the borrowing limit, and lowers job-finding rates. Longer unemployment spells and weaker bargaining power when close to the borrowing constraint depress earnings disproportionately for low-asset workers.

While the cross-sectional nature of this exercise does not capture the same effects as we see in the panel data in the empirical section, this exercise gives us insights into the mechanism following an aggregate productivity shock. The key mechanism is that wage effects persist even after the unemployment rate has returned to its steady-state distribution. A negative TFP shock depletes individual assets in unemployment and pushes more agents toward the borrowing constraint. Since wealth rebuilds only gradually, worker bargaining positions recover more slowly than the unemployment rate itself, and wages remain depressed even after aggregate conditions return to their steady state. This persistence is central to how the model links aggregate shocks to long-lived distributional effects.

3.11. Policy Counterfactual: Elimination of the Unemployment Safety Net

I use the model to conduct a counterfactual policy analysis in which the safety net available during unemployment is sharply reduced. Operationally, I lower the value of home production while unemployed to near zero, setting bu=0.01 for quantitative tractability. All other parameters in this exercise remain at their calibrated values. I then compare the stationary equilibrium of this counterfactual economy to the stationary equilibrium of the baseline calibration. For ease of exposition, I will refer to this experiment as a reduction in the government safety net. Economically, this policy exercise can also be thought of as an environment with no unemployment insurance (UI). Therefore in this policy counterfactual, unemployed workers must self-insure through precautionary savings to smooth consumption during unemployment.

This policy is particularly salient in this paper studying outcomes for new college graduates, who by and large do not not meet work-history requirements to qualify for UI immediately upon graduation. The policy experiment therefore looks at a counterfactual economy in which no one is eligible for unemployment insurance. The lack of government insurance would initially increase the importance of initial assets, which correspond proportionally to parental income in the data.

Figure 8 plots consumption and savings policy functions under the baseline calibration, denoted hereafter for by solid lines, as well as in the counterfactual economy without a government safety net and no unemployment insurance, $b_u = 0.01$

Consumption is lower across the asset distribution in the counterfactual because the

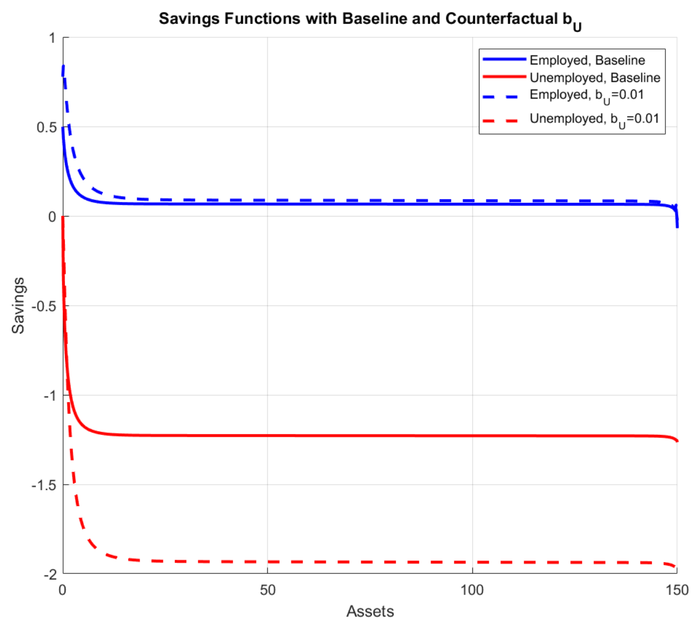


FIGURE 8. Savings policy functions in baseline compared to policy without earnings in unemployment.

Note: Baseline calibrated model shown in solid lines, while dashed lines show counterfactual with no earnings in unemployment. Savings policy functions for employed workers in blue and for unemployed workers in red.

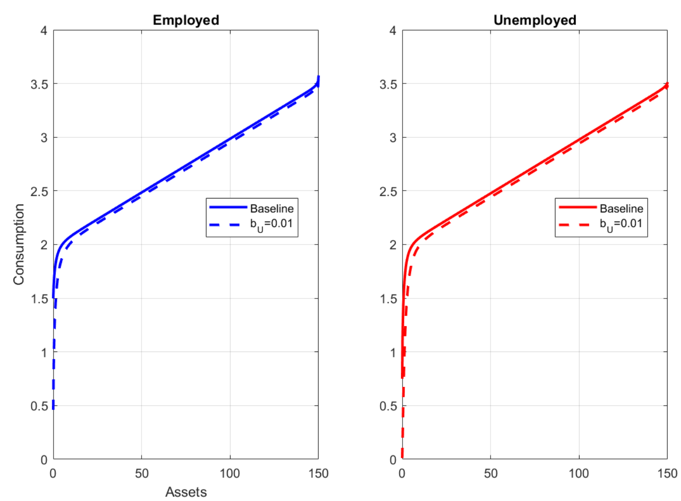


FIGURE 9. Consumption policy function in baseline compared to policy without earnings in unemployment.

Note: Baseline calibrated model shown in solid lines, while dashed lines show counterfactual with no earnings in unemployment. Consumption policy functions for employed workers in left panel in blue and policies for unemployed workers in right panel in red.

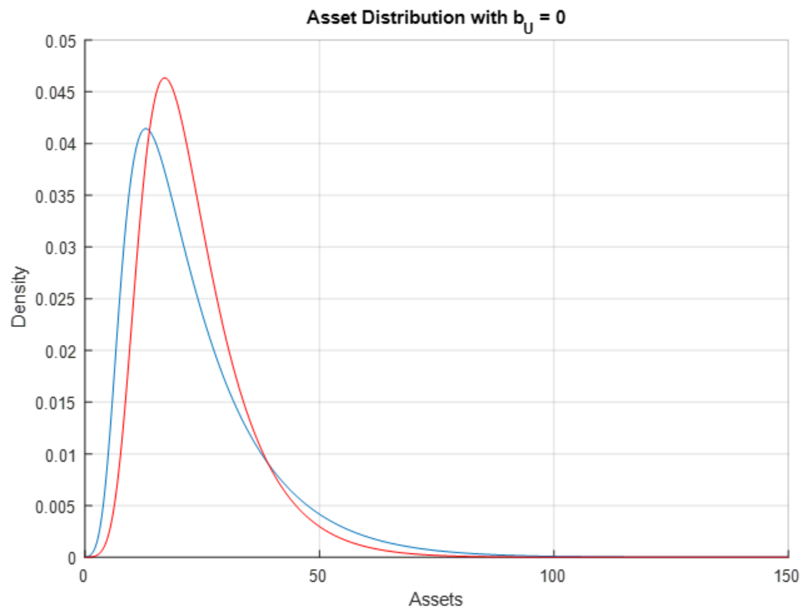


FIGURE 10. Stationary asset distribution in the baseline (blue) and counterfactual with no UI where $b_U \approx 0$ (red)

stronger motive to self-insure when earnings in unemployment approach zero raises desired precautionary savings. While we can see that consumption declines for all levels of assets, the effect is most pronounced at low asset levels where the borrowing constraint is more binding when unemployment income goes to zero. Consistent with this, savings by employed workers increase throughout the distribution, especially near the borrowing constraint. Under the counterfactual policy, there is a larger region of assets at the bottom of the distribution where consumption smoothing is limited, as seen by the more concave shape of the counterfactual policy functions close to the borrowing constraint. For employed individuals, savings increase modestly, but the biggest effect of the counterfactual policy is felt in unemployment. Unemployed workers no longer have any home production or unemployment benefit, and receive virtually no earnings while unemployed. Thus unemployed worker must smooth consumption by dissaving more aggressively than in the baseline upon separation from their job.

Figure 10 compares the stationary asset distribution in the baseline (blue) and counterfactual (red). We observe that the stationary asset distribution becomes more compressed when unemployment income approaches zero. The large decline in earnings for unemployed workers significantly lowers the worker's value in unemployment across the asset distribution, and makes the borrowing constraint more binding for low asset individuals as there is no longer a government safety net they can rely on. Thus, individuals increase their precautionary savings when employed - especially at low asset levels - to be able to smooth against separation shocks. This corresponds to the larger accumulation of assets under this policy in the left-hand tail of the asset distribution compared to the baseline model. On the right tail, however, assets are lower: when separations occur, unemployed workers must finance consumption entirely from their own wealth, leading to larger dissaving spells and slower subsequent re-accumulation relative to the baseline.

Wages also decline across the asset distribution. When wages are determined by bargaining, lowering b_u reduces the unemployed worker's outside option. This weakens the worker's bargaining position and depresses wages across the asset distribution in the policy counterfactual with no government safety net relative to the baseline. As shown in the figure, the wage schedule declines for workers across human capital levels. Again, workers close to the borrowing constraint are much more affected because the borrowing constraint is more binding, and therefore lowers wages disproportionately more for low asset individuals.

I also run a one-time unanticipated 3% drop in productivity with unemployment insurance approaching zero and compare it to the baseline transition dynamics. Similar to the stationary distribution, we see a larger effect for wages on impact at the left-tail of the asset distribution when compared to the baseline model, which is shown in Figure 12.

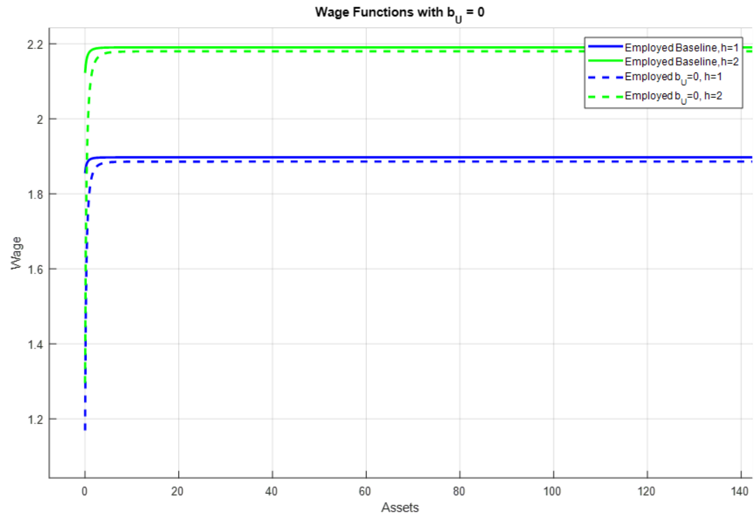


FIGURE 11. Wage Schedule in baseline (solid) and counterfactual (dashed). Green denotes high human capital level, and blue denotes low human capital level

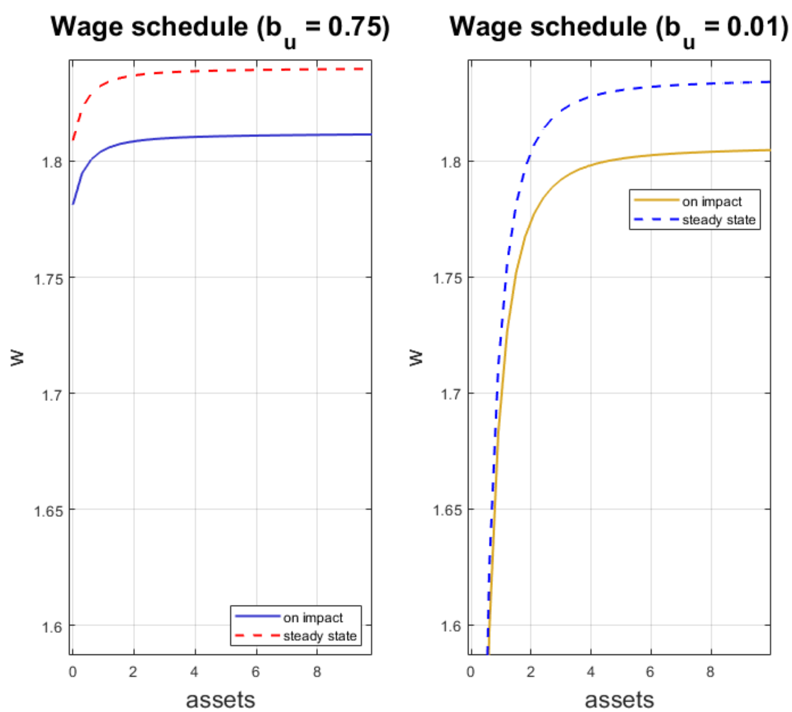


FIGURE 12. Wage schedule on impact following a transitory, unanticipated MIT shock with 3% decrease in $\mathcal{Z}$. Baseline on left panel, policy counterfactual with no unemployment insurance in right panel

4. Conclusion

This paper studies the unequal impact of initial labor market conditions by parental income, and how it can create persistent differences in long-run outcomes. Using large Canadian administrative tax data, I analyze the relationship between parental income and child labor market trajectories, focusing on how aggregate economic conditions influence the wage dynamics of new labor market entrants. I find significant and lasting effects on wages that are unequal by parental income group, with entrants from lower-income families being more vulnerable to downturns and experiencing larger earnings declines when starting their careers during a recession. Motivated by these insights, I develop a model incorporating financial constraints and frictional sector choices within a heterogeneous agent model to quantify the role of initial parental income.

The mechanism I explore is that parental income acts as a safety net, whereby individuals with low-income parents find themselves closer to the borrowing constraint compared to their peers with better-off parents. Individuals with lower-income parents are initially more likely to be in jobs with lower earnings, compared to their peers from higher family income backgrounds. When the initial labor market conditions worsen such as during a recession, this mechanism is amplified as the job finding rate decreases.

My findings help to better understand a key driver of intergenerational inequality and have important policy implications. I suggest that targeted economic policy of unemployment insurance for new labor market entrants can alleviate scarring effects for marginal workers from lower-income families. Such policies are particularly relevant in contexts where inter-generational income and human capital mobility is low.

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Appendix A. Appendix

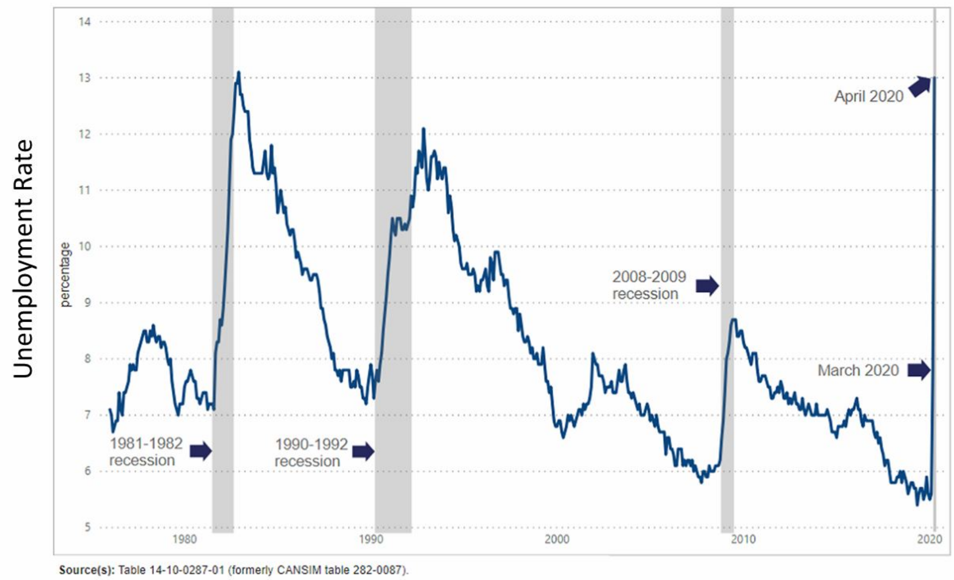


FIGURE A1. Unemployment Rate in Canada, 1975- 2020
Recessions shaded in gray

TABLE A1. Real Labor Earnings Response to 1 p.p. increase in initial unemployment rate UR_{c0}, bottom 20% and top 80% of parental income

	Dependent variable: Real Labor Earnings in 2002 CAD					
	(1)	(2)	(3)	(4)	(5)	(6)
UR Results *j* Years After Graduation						
Reference Group is Bottom 20% of Parental Income						
$\gamma_{0,j=0}$	-2,478.416***	-2,564.204***	-2,533.158***	-2,614.899***	-2,616.457***	-2,601.437***
	(269.735)	(266.568)	(264.923)	(262.212)	(262.215)	(256.766)
$\gamma_{0,j=1}$	-2,358.815***	-2,406.333***	-2,511.994***	-2,660.625***	-2,662.264***	-2,614.033***
	(273.03)	(269.825)	(268.16)	(265.415)	(265.417)	(259.896)
$\gamma_{0,j=2}$	-2,151.935***	-2,155.812***	-2,365.000***	-2,566.850***	-2,568.559***	-2,407.400***
	(273.512)	(270.301)	(268.636)	(265.886)	(265.889)	(260.355)
$\gamma_{0,j=3}$	-2,211.117***	-2,135.527***	-2,128.101***	-2,549.252***	-2,551.209***	-2,273.418***
	(338.807)	(334.83)	(332.763)	(329.363)	(329.366)	(322.506)
$\gamma_{0,j=5}$	-2,337.990***	-2,143.261***	-1,854.507***	-2,522.854***	-2,524.605***	-2,127.655***
	(399.288)	(394.602)	(392.171)	(388.177)	(388.179)	(380.096)
$\gamma_{0,j=8}$	-1,645.904***	-1,570.259***	-1,522.383***	-2,178.521***	-2,180.230***	-1,929.869***
	(468.911)	(463.406)	(460.545)	(455.843)	(455.845)	(446.349)
$\gamma_{0,j=10}$	175.217	243.156	56.43	-588.499	-590.29	-361.678
	(485.019)	(479.325)	(476.368)	(471.5)	(471.502)	(461.681)
UR x Family Income Group Results						
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income						
$\gamma_{p,j=0}$	490.642*	483.262*	540.715**	546.344**	545.698**	440.186*
	(277.879)	(274.617)	(272.922)	(270.128)	(270.128)	(264.503)
$\gamma_{p,j=1}$	910.644***	901.228***	695.978**	692.424**	691.784**	640.292**
	(284.312)	(280.975)	(279.244)	(276.384)	(276.385)	(270.628)
$\gamma_{p,j=2}$	1,010.040***	1,001.086***	828.098***	819.889***	819.264***	752.918***
	(286.629)	(283.264)	(281.518)	(278.635)	(278.636)	(272.832)
$\gamma_{p,j=3}$	676.469*	717.560**	555.255	600.813*	600.115*	617.670*
	(357.404)	(353.209)	(351.03)	(347.436)	(347.436)	(340.199)
$\gamma_{p,j=5}$	-23.355	28.508	-22.738	136.114	135.288	216.641
	(423.72)	(418.746)	(416.161)	(411.9)	(411.901)	(403.319)
$\gamma_{p,j=8}$	-609.917	-547.585	-567.683	-264.921	-265.829	-183.363
	(498.199)	(492.351)	(489.311)	(484.3)	(484.3)	(474.21)
$\gamma_{p,j=10}$	-1,274.259**	-1,222.922**	-1,261.372**	-945.286*	-946.083*	-846.190*
	(515.017)	(508.971)	(505.829)	(500.648)	(500.648)	(490.217)
Age	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y
In School			Y	Y	Y	Y
Degree Type				Y	Y	Y
College or Uni					Y	Y
Major						Y
R^2	0.147	0.167	0.177	0.194	0.194	0.227
Adjusted R^2	0.147	0.167	0.177	0.194	0.194	0.227

Note: * p<0.1; ** p<0.05; *** p<0.01

Note: Estimates show the effect of a 1 p.p. increase in the initial unemployment rate UR_{c0} in graduation year c on real labor earnings for graduates by parental income group. Estimates $\gamma_{0,j}$ correspond to the average effect of the UR at graduation for all individuals j years after graduation in regression specification equation 1, while estimates $\gamma_{p,j}$ correspond to the differential interaction effect of UR_{c0} for graduates from the top 80% of parental income compared to those from the bottom 20%. See text for detailed description of parental income grouping. Real labor earnings are in 2002 CAD and include zero earnings. Individual controls include age, gender, reference year fixed effects,enrolled in post-secondary schooling indicator, post-secondary degree type, two year college or four year university, and major field of study. Standard errors in parentheses. Source: Canadian TIFF–PSIS.

TABLE A2. Real Log Positive Labor Earnings Response to 1 p.p. increase in initial UR_{c0}, bottom 20% and top 80% of parental income

	Dependent variable: Real Positive Log Labor Earnings						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
UR Results j Years After Graduation							
Reference Group is Bottom 20% of Parental Income							
$\gamma_{0,j=0}$	-0.057***	-0.058***	-0.057***	-0.060***	-0.060***	-0.061***	-0.053***
	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=1}$	-0.031***	-0.032***	-0.036***	-0.040***	-0.040***	-0.039***	-0.033***
	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=2}$	0.001	0.001	-0.007	-0.012	-0.012	-0.007	0.001
	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
$\gamma_{0,j=3}$	-0.025**	-0.022**	-0.020*	-0.031***	-0.031***	-0.022**	-0.012
	(0.011)	(0.011)	(0.011)	(0.01)	(0.01)	(0.01)	(0.01)
$\gamma_{0,j=5}$	-0.034***	-0.028**	-0.016	-0.033***	-0.033***	-0.019	-0.009
	(0.013)	(0.013)	(0.013)	(0.012)	(0.012)	(0.012)	(0.012)
$\gamma_{0,j=8}$	-0.013	-0.011	-0.009	-0.028*	-0.028*	-0.02	-0.011
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
$\gamma_{0,j=10}$	0.029*	0.031**	0.024	0.004	0.004	0.01	0.018
	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)
UR x Family Income Group Results							
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income							
$\gamma_{p,j=0}$	0.026***	0.025***	0.028***	0.027***	0.027***	0.022***	0.020**
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
$\gamma_{p,j=1}$	0.032***	0.031***	0.021**	0.019**	0.019**	0.017**	0.019**
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
$\gamma_{p,j=2}$	0.020**	0.020**	0.011	0.01	0.009	0.006	0.007
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)
$\gamma_{p,j=3}$	0.020*	0.021*	0.012	0.013	0.013	0.012	0.014
	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.01)
$\gamma_{p,j=5}$	-0.018	-0.016	-0.019	-0.014	-0.014	-0.012	-0.01
	(0.014)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.012)
$\gamma_{p,j=8}$	-0.024	-0.021	-0.022	-0.012	-0.012	-0.01	-0.008
	(0.016)	(0.016)	(0.016)	(0.015)	(0.015)	(0.015)	(0.014)
$\gamma_{p,j=10}$	-0.026	-0.025	-0.028*	-0.017	-0.017	-0.013	-0.009
	(0.017)	(0.016)	(0.016)	(0.016)	(0.016)	(0.016)	(0.015)
Age	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y
Degree Type				Y	Y	Y	Y
College or Uni					Y	Y	Y
Major						Y	Y
Industry FE Two-Digit NAICS							Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331

Note: * p<0.1; ** p<0.05; *** p<0.01

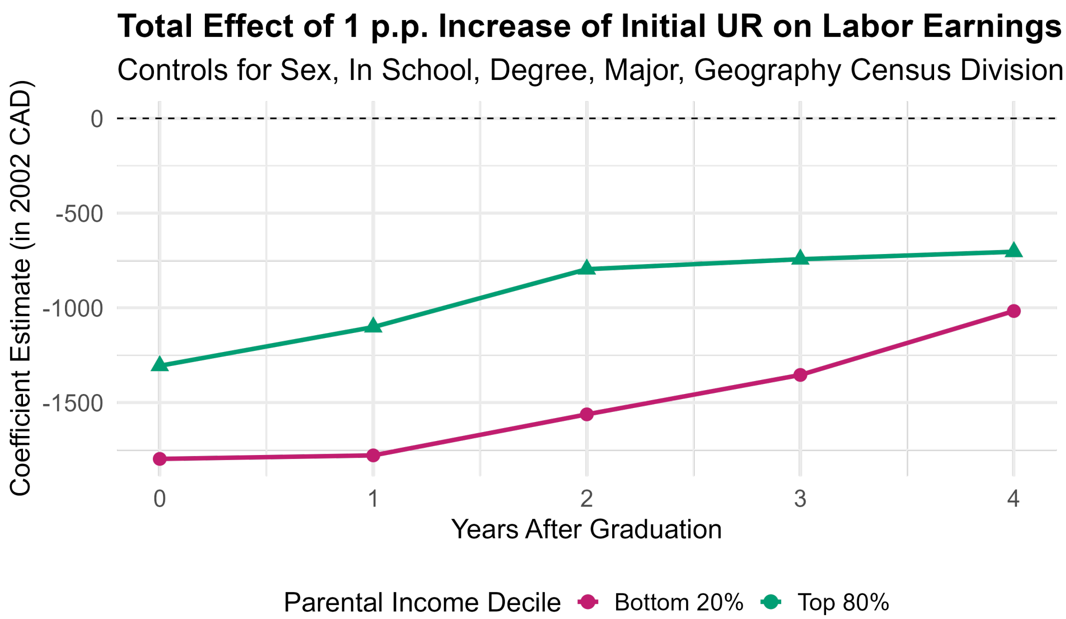
Note: Estimates show the effect of a 1 p.p. increase in the initial unemployment rate UR_{c0} in graduation year c on log positive labor earnings for graduates by parental income group. Estimates $\gamma_{0,j}$ correspond to the average effect of the UR at graduation for all individuals j years after graduation in regression specification equation 1, while estimates $\gamma_{p,j}$ correspond to the differential interaction effect of UR_{c0} for graduates from the top 80% of parental income compared to those from the bottom 20%. See text for detailed description of parental income grouping. Real labor earnings are in 2002 CAD and greater than zero. Individual controls include age, gender, reference year fixed effects, enrolled in post-secondary schooling indicator, post-secondary degree type, two year college or four year university, NAICS industry of employment fixed effects at the two-digit level and major field of study. Standard errors in parentheses. Source: Canadian TIFF–PSIS.

TABLE A3. Real Log Positive Full-Time Labor Earnings Response to 1 p.p. increase in initial UR_{c0}, bottom 20% and top 80% of parental income

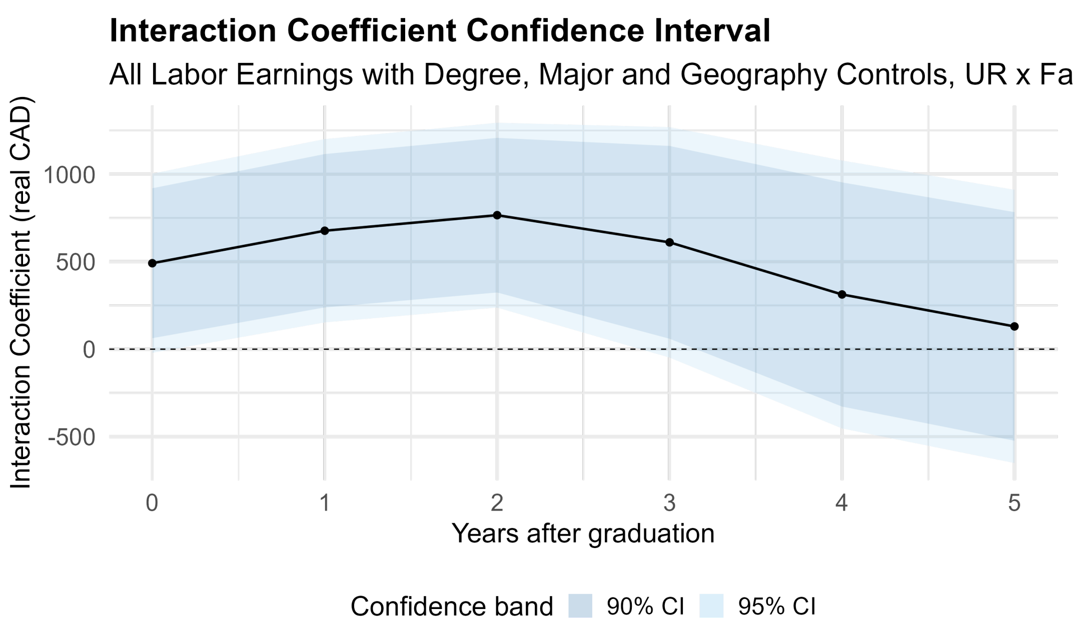
	Dependent variable: Real Positive Log Labor Earnings above 10k						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
UR Results j Years After Graduation							
Reference Group is Bottom 20% of Parental Income							
$\gamma_{0,j=0}$	-0.034***	-0.034***	-0.034***	-0.035***	-0.036***	-0.035***	-0.032***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)
$\gamma_{0,j=1}$	-0.040***	-0.040***	-0.040***	-0.041***	-0.042***	-0.039***	-0.033***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{0,j=2}$	-0.039***	-0.038***	-0.040***	-0.044***	-0.045***	-0.040***	-0.030***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{0,j=3}$	-0.034***	-0.032***	-0.031***	-0.041***	-0.042***	-0.036***	-0.026***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{0,j=5}$	-0.030***	-0.026***	-0.022***	-0.038***	-0.038***	-0.030***	-0.023***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{0,j=8}$	-0.013	-0.011	-0.009	-0.028*	-0.028*	-0.02	-0.011
	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)	(0.014)
$\gamma_{0,j=10}$	-0.003	-0.001	-0.003	-0.019**	-0.019**	-0.012	-0.003
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
UR x Family Income Group Results							
Interaction of UR_{c0} x Family Income Group p: Results for Top 80% of Parental Income							
$\gamma_{p,j=0}$	0.012*	0.011*	0.012*	0.014**	0.014**	0.015**	0.017***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{p,j=1}$	0.015***	0.016***	0.012**	0.012**	0.012**	0.012**	0.013***
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{p,j=2}$	0.017***	0.016***	0.014***	0.014***	0.014***	0.014***	0.012**
	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
$\gamma_{p,j=3}$	0.006	0.007	0.005	0.005	0.005	0.008	0.007
	(0.007)	(0.007)	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)
$\gamma_{p,j=5}$	-0.007	-0.007	-0.007	-0.005	-0.005	-0.002	-0.001
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)
$\gamma_{p,j=8}$	-0.021**	-0.020**	-0.020**	-0.013	-0.013	-0.01	-0.011
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)	(0.008)
$\gamma_{p,j=10}$	-0.021**	-0.020**	-0.021**	-0.013	-0.013	-0.011	-0.011
	(0.010)	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.008)
Age	Y	Y	Y	Y	Y	Y	Y
Refyear FE	Y	Y	Y	Y	Y	Y	Y
Sex		Y	Y	Y	Y	Y	Y
In School			Y	Y	Y	Y	Y
Degree Type				Y	Y	Y	Y
College or Uni					Y	Y	Y
Major						Y	Y
Industry FE Two-Digit NAICS							Y
R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331
Adjusted R^2	0.184	0.198	0.22	0.238	0.238	0.276	0.331

Note: * p<0.1; ** p<0.05; *** p<0.01

Note: Estimates show the effect of a 1 p.p. increase in the initial unemployment rate UR_{c0} in graduation year c on log positive full-time labor earnings by parental income group. Estimates $\gamma_{0,j}$ correspond to the average effect of the UR at graduation for all individuals j years after graduation in regression specification equation 1, while estimates $\gamma_{p,j}$ correspond to the differential interaction effect of UR_{c0} for graduates from the top 80% of parental income compared to those from the bottom 20%. See text for detailed description of parental income grouping. Real positive labor earnings are in 2002 CAD. Individual controls include age, gender, reference year fixed effects, enrolled in post-secondary schooling indicator, post-secondary degree type, two year college or four year university, NAICS industry of employment fixed effects at the two-digit level and major field of study. Standard errors in parentheses. Source: Canadian TIFF–PSIS.

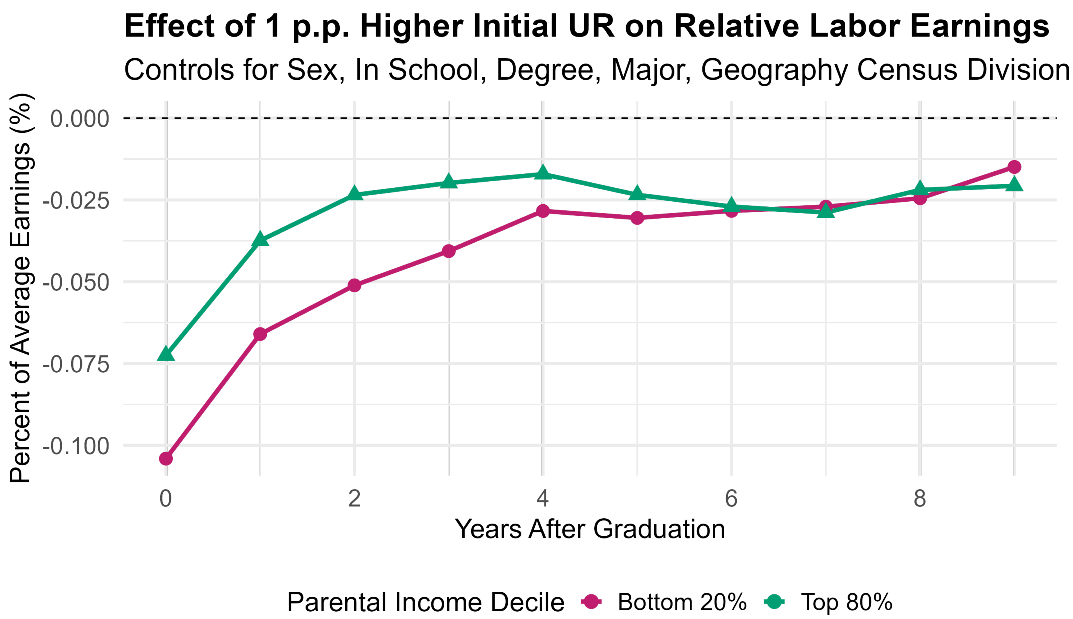


A. Level effect on average earnings by parental income group with a one percentage point increase in the unemployment rate at graduation, from zero to four years after graduation. Conditioning on age, gender, year fixed effects, post-secondary degree, institution type, major, geography census division. All levels of earnings including zero earnings. Source: PSIS-T1FF

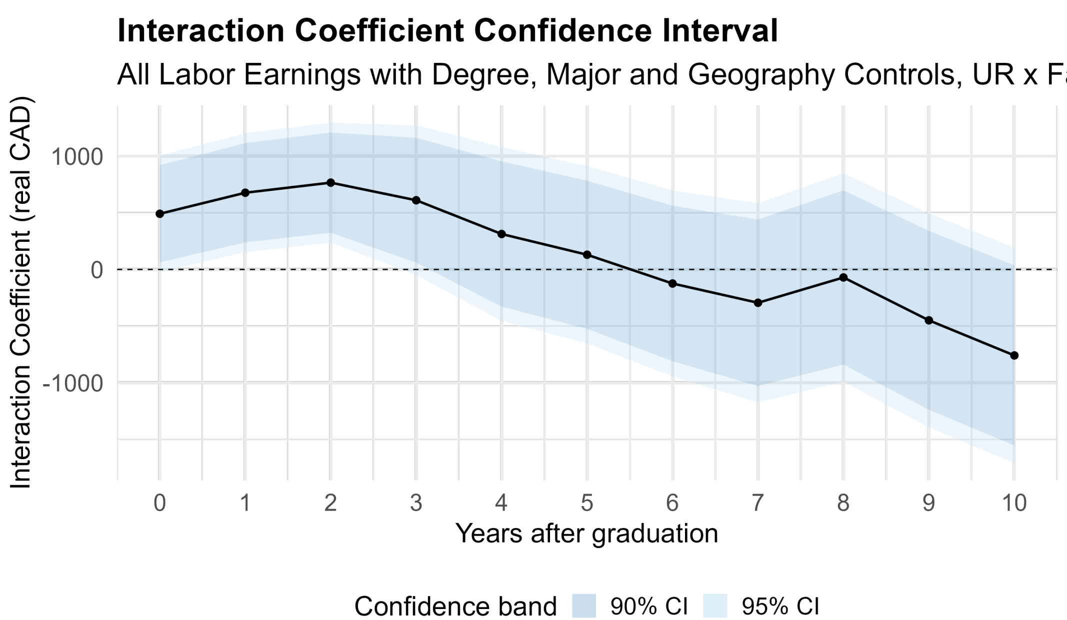


B. Confidence intervals of interaction term of parental income group and effect of initial unemployment rate at graduation.

FIGURE A2. Differential effect of initial unemployment rate by parental income group



A. Relative effect on average earnings by parental income group with a one percentage point increase in the unemployment rate at graduation in the years after graduation, relative to own parental group average earnings. Conditioning on age, gender, year fixed effects, post-secondary degree, institution type, major, geography census division. All levels of earnings including zero earnings. Source: PSIS-T1FF



B. Confidence intervals of interaction term of parental income group and effect of initial unemployment rate at graduation.

FIGURE A3. Differential effect of initial unemployment rate by parental income group